

A REPORT

ON

REAL-TIME THREAT DETECTION AND INCIDENT ANALYSIS

USING ENDPOINT TELEMETRY IN ENTERPRISE ENVIRONMENTS

BY

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AT

Industry-Sponsored Project

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This project, titled “**Real-Time Threat Detection and Incident Analysis Using Endpoint Telemetry in Enterprise Environments**,” provided me with a valuable opportunity to apply theoretical concepts to practical cybersecurity scenarios using technologies such as Microsoft Sentinel, Azure Log Analytics, Sysmon, and endpoint telemetry data collection. The knowledge and experience gained through this work have significantly contributed to my professional and academic growth.

Finally, I would like to express my heartfelt appreciation to my family and friends for their unwavering support, encouragement, patience, and motivation throughout my academic journey and during the successful completion of this dissertation.

Key Words: Endpoint Telemetry, Microsoft Sentinel, Sysmon, SIEM, Threat Detection, Incident Analysis, Cybersecurity, Log Analytics Workspace, Azure Monitor Agent, Kusto Query Language (KQL).

Project Areas: Cybersecurity.

Abstract: Cybersecurity has become an essential part of modern enterprise environments due to the increasing number of cyber threats and attacks. Organizations today rely heavily on digital systems, and any security breach can lead to data loss, financial impact, and operational disruption. As a result, continuous monitoring and protection of IT infrastructure has become very important. One of the key areas in cybersecurity is endpoint security. Endpoint devices such as desktops, laptops, and servers are widely used in organizations and act as major entry points for cyber-attacks. These systems generate large amounts of data in the form of logs, known as endpoint telemetry, which can be analysed to understand system behavior and detect potential threats. Real-time threat detection plays an important role in identifying suspicious activities at an early stage and preventing security incidents. By analyzing endpoint telemetry using tools like SIEM, it becomes possible to detect abnormal activities, generate alerts, and perform incident analysis effectively. This project focuses on using endpoint telemetry to simulate a real-world security monitoring environment and understand how threats are detected and managed in enterprise systems. In modern organizations, endpoints such as desktops, laptops, and servers act as primary entry points for cyber-attacks. Traditional security solutions alone are often insufficient for detecting advanced threats. Therefore, organizations increasingly rely on endpoint telemetry and Security Information and Event Management (SIEM) platforms to monitor system activities, detect suspicious behavior, generate alerts, and support structured incident investigations.

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Chapter 1

INTRODUCTION

1.1 Background of the Study.

Cybersecurity has become a critical requirement for modern organizations due to the increasing frequency and sophistication of cyber threats. Enterprises today rely heavily on digital infrastructure, cloud services, and connected devices to support business operations. As a result, any security breach can lead to financial losses, operational disruption, reputational damage, and unauthorized access to sensitive information. To reduce these risks, organizations implement various security controls and monitoring solutions to identify and respond to potential threats at an early stage [7] [8].

One of the most important areas of cybersecurity is endpoint security. Endpoints such as desktops, laptops, servers, and virtual machines are widely used within enterprise environments and often serve as primary targets for attackers. Malicious activities such as unauthorized logins, malware execution, privilege escalation, and suspicious process creation frequently originate at the endpoint level. Therefore, monitoring endpoint activity has become essential for maintaining organizational security [1] [2].

Modern endpoint systems continuously generate large volumes of security-related data in the form of system logs, process information, network events, user activity records, and file operation logs. This data is commonly referred to as endpoint telemetry. Endpoint telemetry provides valuable visibility into the behavior of systems and users, enabling security teams to detect suspicious activities and investigate security incidents more effectively [1] [6].

The growing adoption of Security Information and Event Management (SIEM) platforms has improved the ability of organizations to collect, correlate, and analyze telemetry data from multiple sources. SIEM solutions provide centralized monitoring, alert generation, and incident investigation capabilities, helping security teams identify potential threats in near real-time. In addition, tools such as Sysmon provide detailed endpoint visibility by recording process creation events, network connections, file activities, and other security-relevant actions.

This dissertation focuses on understanding how endpoint telemetry can be collected and utilized for real-time threat detection and incident analysis in an enterprise-like environment. A practical monitoring environment has been developed using Microsoft Azure, Sysmon, Azure Monitor Agent, Data Collection Rules, Log Analytics Workspace, and Microsoft Sentinel. Through the implementation of detection rules and the analysis of security events, the project demonstrates how suspicious activities can be identified, monitored, and investigated within a controlled environment [3] [10] [12].

1.2 Problem Statement.

Enterprise environments generate a large volume of security events on a daily basis. Monitoring these events manually is a challenging task due to the scale of data, the complexity of modern attacks, and the limited resources available to security teams. Traditional security approaches often rely on signature-based detection methods, which may not be sufficient to identify emerging threats, insider activities, or behavior-based attacks.

Security analysts frequently face challenges such as alert fatigue, false positives, delayed threat identification, and limited endpoint visibility. These challenges make it difficult to distinguish genuine threats from normal system activities. As organizations continue to expand their digital infrastructure, there is a growing need for solutions that provide continuous monitoring and enable rapid detection of suspicious activities.

The problem addressed in this dissertation is how endpoint telemetry can be effectively utilized to improve threat detection and support incident analysis through centralized monitoring and security analytics. The study aims to examine how endpoint-generated logs can be collected, analyzed, and converted into meaningful alerts and incidents that assist security teams in identifying potential security threats.

1.3 Research Gap.

A review of existing literature indicates that endpoint monitoring and SIEM technologies are widely used for cybersecurity operations. However, many organizations still face difficulties in effectively utilizing the large volume of endpoint telemetry generated by modern systems. Although various detection solutions are available, challenges related to false positives, delayed response, limited visibility, and efficient incident investigation continue to exist.

Several studies focus on theoretical security monitoring approaches, while practical demonstrations using endpoint telemetry within a controlled enterprise-like environment receive comparatively less attention. Additionally, there is a need to understand how security monitoring tools such as Microsoft Sentinel can be integrated with endpoint telemetry sources to provide meaningful threat detection and incident analysis capabilities.

This research attempts to bridge this gap by implementing a practical endpoint monitoring solution and evaluating how endpoint-generated telemetry can be used to detect suspicious activities and support incident investigation workflows.

1.4 Aim of the Study.

The aim of this dissertation is to study and implement a real-time threat detection and incident analysis framework using endpoint telemetry collected from a Windows-based endpoint environment. The research focuses on understanding how security events can be collected, analyzed, and utilized for identifying suspicious activities through centralized monitoring and security analytics.

1.5 Objectives of Project.

The primary objectives of this dissertation are:

- To study the concept of endpoint telemetry and its role in enterprise security monitoring.
- To configure a Windows-based endpoint environment capable of generating security telemetry.
- To deploy Sysmon for enhanced telemetry collection and visibility.
- To configure Microsoft Sentinel for centralized log monitoring and analysis.
- To collect and analyze security events using Kusto Query Language (KQL).
- To develop detection use cases for identifying suspicious activities.
- To generate alerts and incidents based on selected security scenarios.
- To perform incident analysis using collected telemetry data.
- To evaluate the effectiveness of endpoint telemetry for supporting threat detection activities.

1.6 Scope of the Study.

The scope of this dissertation is limited to a controlled laboratory environment developed within Microsoft Azure. The study primarily focuses on Windows Endpoint monitoring using Sysmon, Windows Security Logs, Azure Monitor Agent, Data Collection Rules, Log Analytics Workspace, and Microsoft Sentinel.

The project includes the collection and analysis of endpoint-generated telemetry, development of detection use cases, alert generation, and incident analysis. The implementation is intended for educational and research purposes and does not attempt to fully replicate the scale and complexity of a large enterprise Security Operations Center (SOC).

1.7 Significance of the Study.

The significance of this study lies in demonstrating how endpoint telemetry can be leveraged to improve security monitoring capabilities within enterprise environments. By utilizing centralized log collection and SIEM-based analysis, organizations can gain better visibility into security-related activities occurring across endpoint systems.

The findings of this dissertation contribute to a better understanding of how telemetry-driven monitoring can assist in detecting suspicious behavior, generating timely alerts, and supporting incident investigation. The study also provides practical insights into the implementation of Microsoft Sentinel-based monitoring solutions, making it relevant for students, researchers, and cybersecurity professionals interested in security operations and threat detection.

Chapter 2

LITERATURE REVIEW

2.1 Introduction.

The rapid growth of digital transformation and interconnected enterprise systems has significantly increased the attack surface available to cyber adversaries. Modern organizations rely heavily on endpoint devices such as workstations, servers, and laptops for day-to-day business operations. These endpoints generate a large volume of security-relevant events that can provide valuable insights into system activities and potential security threats. As cyberattacks continue to evolve in sophistication, organizations require advanced monitoring and detection mechanisms capable of identifying malicious activities in real time.

Endpoint telemetry has emerged as a critical source of security data because it captures detailed information about processes, users, network connections, file operations, and system changes occurring on individual endpoints. When combined with Security Information and Event Management (SIEM) platforms, endpoint telemetry enables centralized monitoring, threat detection, and incident investigation. This chapter reviews the existing literature related to cybersecurity threats, endpoint telemetry, Sysmon, SIEM platforms, threat detection methodologies, and real-time security monitoring.

2.2 Cyber Threat Landscape in Enterprise Environments.

Enterprise environments face a wide range of cybersecurity threats that can disrupt business operations, compromise sensitive information, and cause significant financial losses. Modern attack techniques often target endpoint devices because they serve as entry points into organizational networks. Threat actors exploit vulnerabilities, misconfigurations, and human errors to gain unauthorized access to critical systems [7] [8].

Among the most prevalent threats are malware, ransomware, phishing attacks, credential theft, insider threats, and Advanced Persistent Threats (APTs). Ransomware attacks encrypt organizational data and demand payment for restoration, while phishing campaigns deceive users into disclosing credentials or executing malicious files. Insider threats originate from individuals within the organization who intentionally or unintentionally compromise security controls.

Recent studies suggest that traditional security solutions alone are insufficient for detecting sophisticated attacks. Consequently, organizations increasingly adopt advanced monitoring solutions capable of collecting detailed endpoint activity and correlating security events across multiple systems. Continuous monitoring of endpoint behavior enables security teams to identify indicators of compromise, detect suspicious activities, and respond to incidents more effectively.

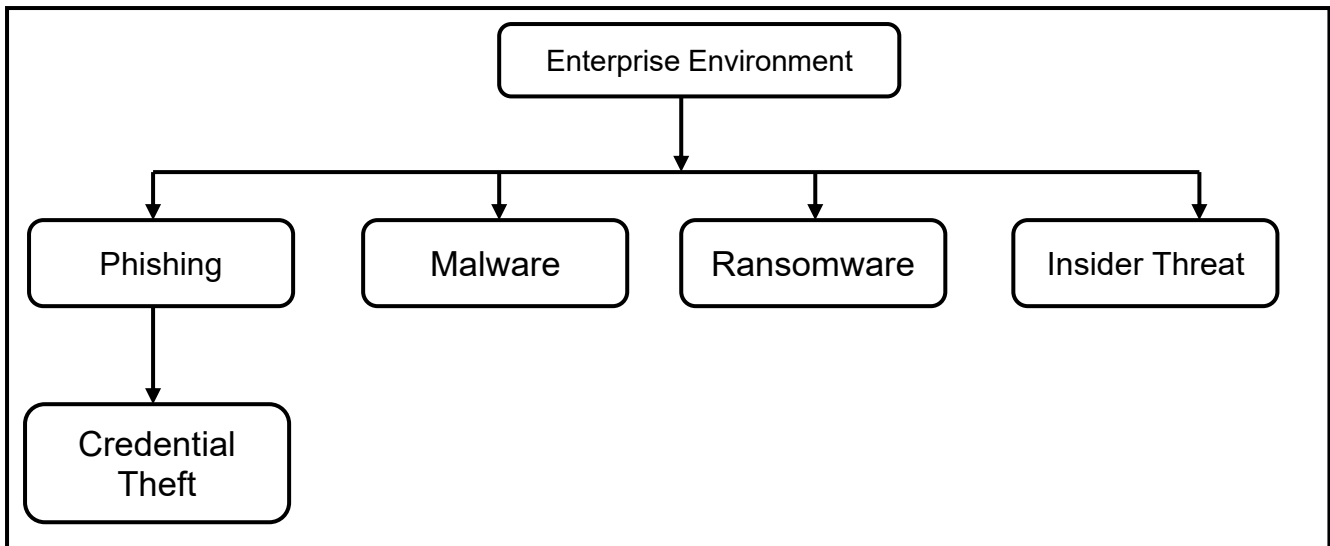


Figure 2.1: Common Cybersecurity Threats in Enterprise Environments.

2.3 Endpoint Telemetry and Security Monitoring.

Endpoint telemetry refers to the collection of detailed operational and security-related information from endpoint devices. Telemetry data provides visibility into system behavior and supports the identification of anomalous or malicious activities. Common sources of endpoint telemetry include operating system logs, application logs, process execution records, network activity logs, registry modifications, and authentication events.

In cybersecurity operations, telemetry serves as the foundation for real-time threat detection. Security analysts use telemetry data to reconstruct attack timelines, investigate incidents, and monitor system health. Unlike traditional log collection methods that focus on limited security events, endpoint telemetry captures extensive details about user and system activities.

The effectiveness of endpoint telemetry depends on the quality, granularity, and reliability of collected data. Rich telemetry enables organizations to detect complex attack behaviors such as privilege escalation, persistence mechanisms, lateral movement, and unauthorized access attempts. As a result, endpoint telemetry has become a critical component of modern Security Operations Centers (SOCs) [1] [5] [6].

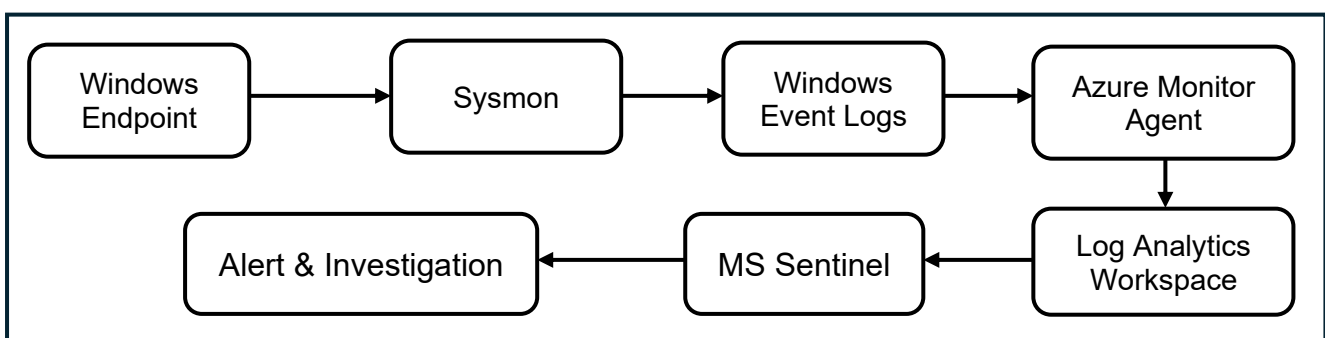


Figure 2.2: Endpoint Telemetry Collection and Monitoring Architecture.

2.4 Endpoint Detection and Response (EDR).

Endpoint Detection and Response (EDR) is a cybersecurity technology designed to continuously monitor endpoint devices for suspicious activities and security incidents. Unlike conventional antivirus solutions that primarily rely on signature-based detection, EDR platforms provide advanced capabilities including behavioral analysis, threat hunting, forensic investigation, and automated response actions.

EDR systems collect endpoint telemetry data and apply analytical techniques to identify indicators of malicious behavior. These solutions often maintain historical records of system activities, enabling investigators to reconstruct attack sequences and understand the progression of security incidents.

The adoption of EDR technologies has become increasingly important as modern cyberattacks frequently utilize fileless malware, PowerShell scripts, and legitimate system tools to evade traditional security controls. EDR solutions improve organizational visibility and facilitate faster detection and containment of cyber threats [1] [2].

2.5 Security Information and Event Management (SIEM).

Security Information and Event Management (SIEM) systems provide centralized platforms for collecting, storing, analyzing, and correlating security logs from multiple data sources. SIEM technologies play a vital role in modern cybersecurity operations by enabling real-time visibility into organizational security events.

A SIEM platform aggregates logs from endpoints, servers, firewalls, security appliances, cloud environments, and network devices. Advanced correlation mechanisms identify relationships between seemingly unrelated events and generate alerts when suspicious patterns are detected.

The ability to centralize monitoring activities significantly improves incident response efficiency. Security analysts can investigate alerts, review historical logs, perform threat hunting activities, and generate compliance reports using a single platform. SIEM solutions also facilitate automation of repetitive security tasks, reducing operational workload and accelerating threat detection [3] [5] [10].

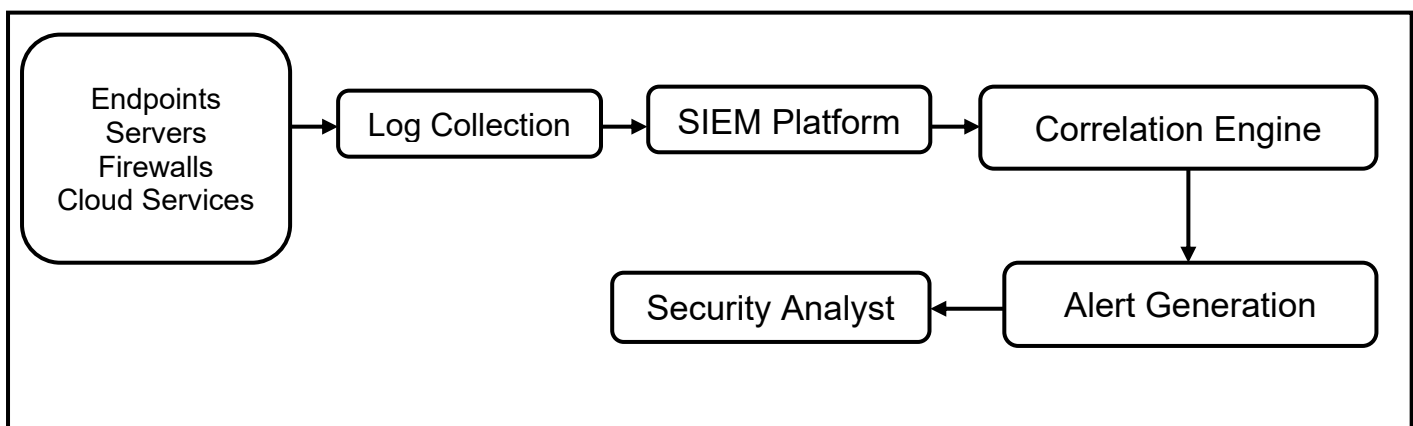


Figure 2.3: SIEM Workflow Architecture.

2.6 Sysmon for Endpoint Monitoring.

System Monitor (Sysmon) is a Windows system service and device driver that provides enhanced visibility into endpoint activities [12]. Developed as part of the Microsoft Sysinternals Suite, Sysmon records detailed information about system events and stores them within the Windows Event Log.

Sysmon significantly extends native Windows logging capabilities by capturing detailed events related to process creation, network connections, file creation, image loading, driver activity, registry modifications, and DNS queries. These events provide security analysts with valuable information for identifying suspicious behaviors and investigating security incidents.

Several event identifiers generated by Sysmon are particularly important from a cybersecurity perspective. Process creation events help identify command execution activities, network connection events reveal communication between systems, and registry modification events can indicate persistence mechanisms used by attackers. Due to its detailed logging capabilities, Sysmon is widely utilized in security monitoring environments and threat detection architectures [12].

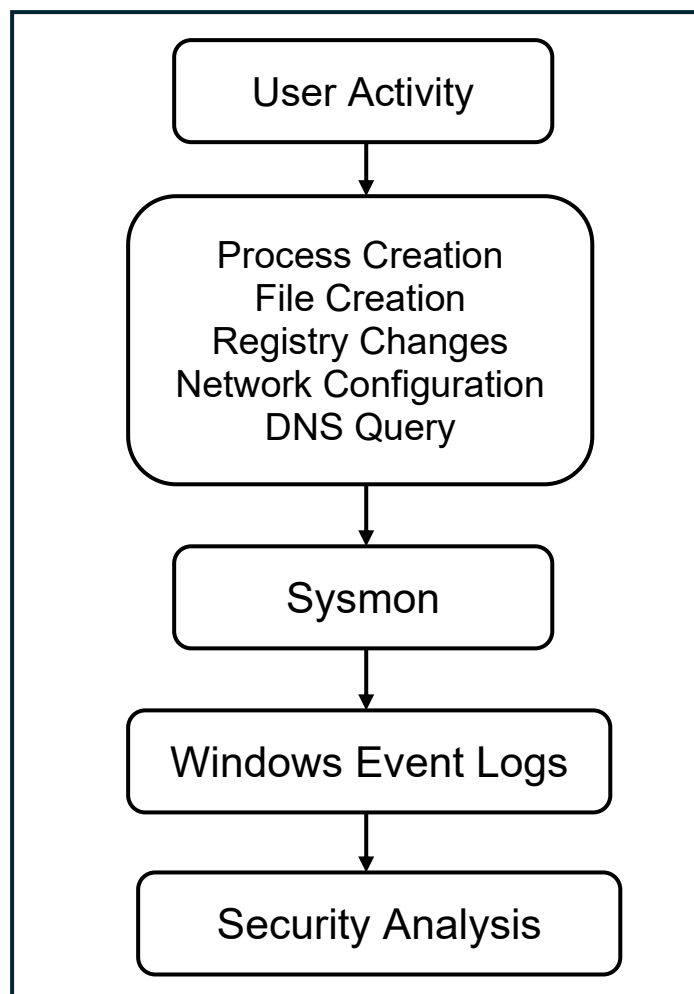


Figure 2.4: Sysmon Event Generation and Logging Workflow.

2.7 Microsoft Sentinel as a Cloud-Native SIEM & SOAR.

Microsoft Sentinel is a cloud-native Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) solution developed specifically for modern enterprise environments. It enables organizations to collect and analyze security data from diverse sources while leveraging cloud-scale analytics and machine learning capabilities [3] [10] [13].

Microsoft Sentinel integrates with Azure services, Windows endpoints, cloud applications, and third-party security products. Through Log Analytics Workspaces, security data can be stored, queried, and visualized efficiently. The platform supports advanced detection mechanisms using Kusto Query Language (KQL), enabling analysts to define custom detection rules based on organizational requirements.

The cloud-native architecture of Microsoft Sentinel improves scalability, accessibility, and operational flexibility while reducing infrastructure management complexities associated with traditional SIEM deployments [3] [10].

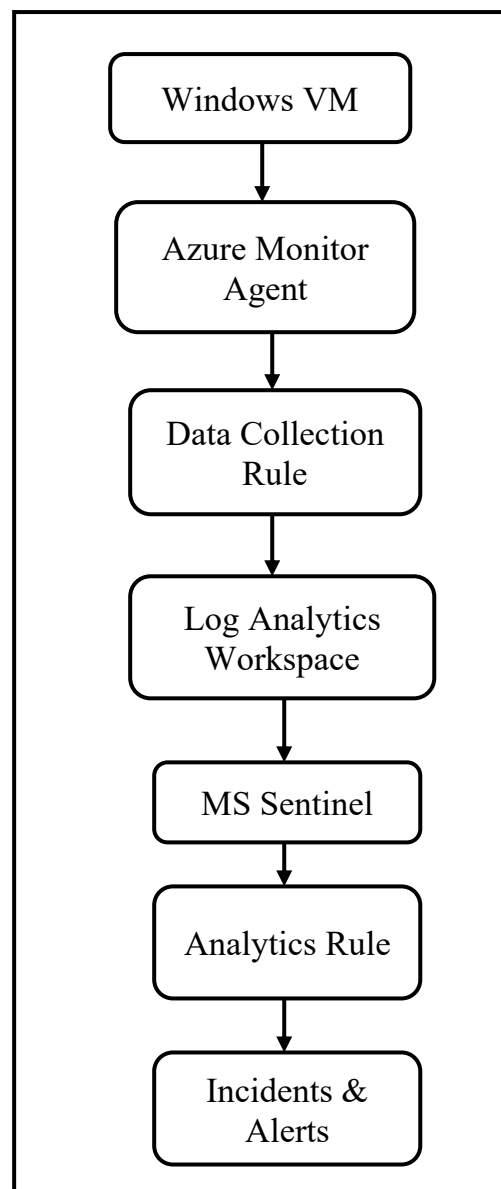


Figure 2.5: Microsoft Sentinel Architecture Used in the Project.

2.8 Threat Detection Techniques:

Cybersecurity monitoring solutions employ multiple detection methodologies to identify malicious activities within enterprise environments.

Signature-Based Detection:

Signature-based detection identifies threats by comparing observed activities against known attack patterns and predefined indicators of compromise. While highly effective against known threats, it is less capable of identifying previously unseen attack techniques.

Anomaly-Based Detection:

Anomaly-based detection establishes a baseline of normal system behavior and flags deviations that may indicate malicious activity. This approach is particularly useful for detecting unknown threats and insider attacks.

Behavior-Based Detection:

Behavior-based detection focuses on identifying malicious sequences of actions rather than individual events. Security solutions analyze user and system behaviors to detect activities that resemble attack techniques commonly used by threat actors.

Rule-Based Detection Using KQL:

Rule-based detection utilizes predefined logic and analytical queries to identify suspicious activities within collected telemetry. Kusto Query Language (KQL) enables analysts to develop customized detection rules for identifying activities such as PowerShell execution, failed login attempts, suspicious process creation, and abnormal network communication.

Detection Type	Strength	Limitation
Signature-Based	Fast Detection	Only Known Threats
Anomaly-Based	Detect Unknown Threats	False Positives
Behavior-Based	Detect Advanced Attacks	Computational Cost
Rule-Based (KQL)	Custom Detection	Requires Fine Tuning

Table 2.1: Comparison of Threat Detection Techniques.

2.9 MITRE ATT&CK Framework.

The MITRE ATT&CK Framework is a globally recognized knowledge base that documents adversarial tactics and techniques observed during real-world cyberattacks. The framework provides a structured methodology for understanding attacker behavior and improving defensive strategies.

MITRE ATT&CK categorizes attack activities into tactics representing attacker objectives and techniques describing specific methods used to achieve those objectives. Security teams use the framework to assess detection coverage, perform threat hunting, and map observed events to known attack behaviors.

By aligning endpoint telemetry with MITRE ATT&CK techniques, organizations can

improve threat visibility and better understand the significance of detected security events. The framework also facilitates communication among security professionals by providing a common language for discussing cyber threats [11] [17].

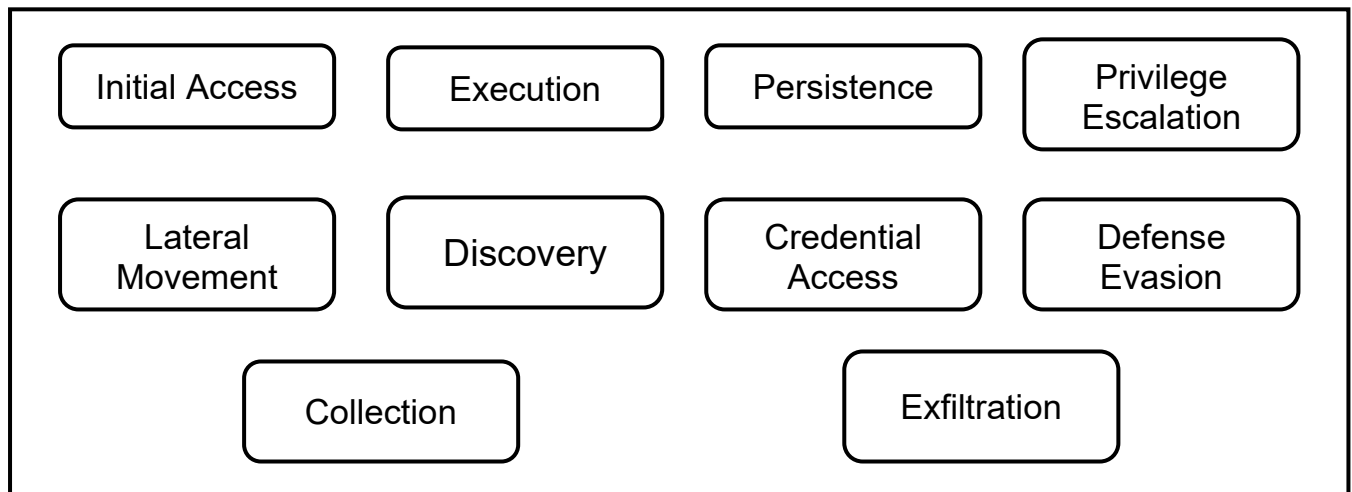


Figure 2.6: Simplified MITRE ATT&CK Attack Lifecycle.

2.10 Summary.

This chapter reviewed the existing literature related to enterprise cybersecurity threats, endpoint telemetry, Endpoint Detection and Response technologies, Sysmon monitoring, SIEM platforms, Microsoft Sentinel, threat detection methodologies, and the MITRE ATT&CK framework. The review highlights the growing importance of endpoint visibility and centralized security analytics in modern enterprise environments. Furthermore, the identified research gap establishes the need for a practical implementation capable of leveraging endpoint telemetry and SIEM technologies for effective real-time threat detection and incident analysis.

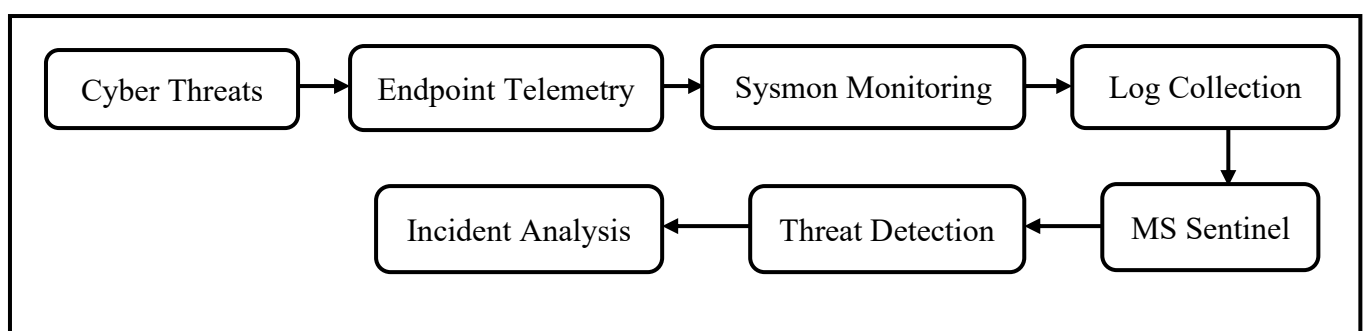


Figure 2.7: Conceptual Relationship Between Endpoint Telemetry, Detection, and Incident Analysis.

Chapter 3

SYSTEM DESIGN AND METHODOLOGY

3.1 Introduction.

This chapter explains the design methodology adopted for developing a real-time threat detection and incident analysis framework using endpoint telemetry. The main purpose of this chapter is to describe how the proposed system was planned before implementation, including the architecture, data flow, telemetry collection strategy, detection logic, and incident analysis workflow.

The system was designed to simulate an enterprise security monitoring environment in a controlled lab setup. Endpoint activity was selected as the primary source of telemetry because endpoints often act as the initial point of compromise in enterprise networks. The proposed methodology focuses on collecting endpoint-level events, forwarding them to a centralized analytics platform, applying detection logic, and supporting incident investigation through a SIEM-based workflow.

Unlike traditional endpoint monitoring approaches that only focus on local logs, the proposed design integrates endpoint telemetry with cloud-based monitoring and SIEM capabilities. This design supports improved visibility, centralized analysis, and faster identification of suspicious activity.

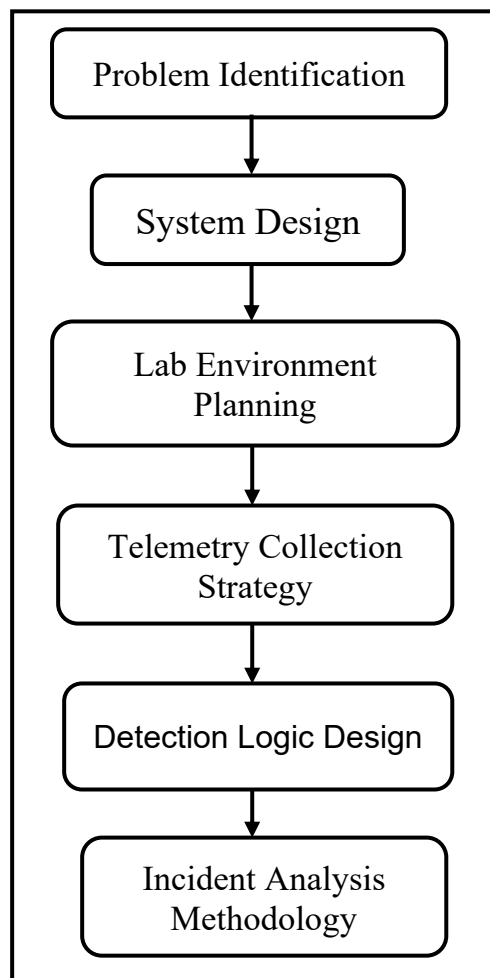


Figure 3.1: Methodology Overview.

3.2 Design Objectives of the Proposed System.

The proposed system was designed with the objective of demonstrating how endpoint telemetry can support real-time threat detection and incident analysis in an enterprise-like environment. The design focuses on creating a complete monitoring pipeline, starting from endpoint activity generation and ending with SIEM-based alert investigation.

The main design objectives are as follows:

1. To collect detailed endpoint telemetry from a Windows-based system.
2. To monitor process activity, network communication, authentication events, and system-level changes.
3. To forward endpoint logs to a centralized monitoring workspace.
4. To analyze collected telemetry using query-based detection logic.
5. To generate alerts for suspicious activities.
6. To support structured incident analysis using Microsoft Sentinel.
7. To evaluate whether endpoint telemetry improves visibility during threat investigation.

This objective-based design ensures that the project is not limited to tool installation but also demonstrates the complete security monitoring lifecycle.

Design Objective	Expected Outcome
Endpoint telemetry collection	Visibility into endpoint activity
Centralized log storage	Availability of logs for analysis
SIEM integration	Correlation and monitoring of suspicious events
KQL-based detection	Identification of selected suspicious behaviors
Alert generation	Automated notification of detected activity
Incident analysis	Structured investigation of security events

Table 3.1: Design Objectives and Expected Outcomes.

3.3 Proposed System Architecture.

The proposed architecture follows a layered security monitoring model. Each layer performs a specific role in the detection and analysis pipeline. This layered approach improves clarity, simplifies troubleshooting, and makes the system easier to expand in future work.

The architecture consists of the following layers:

Endpoint Layer:

This layer represents the Windows virtual machine used to generate endpoint activity. Activities such as process execution, command-line usage, file creation, login attempts, and network communication originate from this layer.

Telemetry Generation Layer:

Sysmon is used to generate detailed telemetry from endpoint activities. It provides more granular visibility than default Windows logs by recording process creation, network connections, file operations, registry changes, and DNS queries.

Collection Layer:

Azure Monitor Agent and Data Collection Rules are used to collect selected logs from the endpoint and forward them to Log Analytics Workspace.

Storage and Query Layer:

Log Analytics Workspace stores the collected telemetry and provides query capabilities using Kusto Query Language.

Detection and Analysis Layer:

Microsoft Sentinel is used as the SIEM platform for analytics rules, alert generation, incident management, and investigation.

Microsoft's documentation describes Microsoft Sentinel as providing attack detection, threat visibility, proactive hunting, and threat response capabilities. This supports its use as the SIEM component in the proposed design [10].

Detection and Analysis Layer	- Microsoft Sentinel, Analytics Rules, Incidents
Storage and Query Layer	- Log Analytics Workspace, KQL
Collection Layer	- Azure Monitor Agent, Data Collection Rule
Telemetry Generation Layer	- Sysmon, Windows Event Logs
Endpoint Layer	- Windows Virtual Machine

Table 3.2: Layered Architecture of the Proposed System.

3.4 Component Selection and Justification.

The components used in the proposed system were selected based on their relevance to endpoint monitoring, SIEM integration, and real-time analysis. Each component contributes to a specific stage of the security monitoring workflow.

A Windows virtual machine was selected as the endpoint because it provides a suitable environment for generating endpoint telemetry and testing monitoring configurations. Sysmon was selected because it provides detailed endpoint-level visibility, including process creation, network connections, file creation, registry activity, and DNS events. Azure Monitor Agent was selected to collect endpoint logs and forward them to Azure. Log Analytics Workspace was selected as the centralized log storage and querying platform. Microsoft Sentinel was selected as the SIEM solution because it supports security monitoring, analytics rules, investigations, and incident management.

This component selection provides a practical and scalable design for implementing a telemetry-driven detection framework [3] [10] [12]. The project remains within a controlled lab environment and does not require production or client data, which supports ethical and safe experimentation.

Component	Purpose in Design
Windows Virtual Machine	Simulated enterprise endpoint
Sysmon	Detailed endpoint telemetry generation
Windows Event Logs	Local log storage source
Azure Monitor Agent	Forwarding logs to Azure
Data Collection Rule	Defining what telemetry should be collected
Log Analytics Workspace	Centralized log storage and querying
Microsoft Sentinel	SIEM-based detection and incident analysis
KQL	Query-based detection logic

Table 3.3: Components Used in the Proposed Design.

3.5 Telemetry Collection Design.

The telemetry collection design defines what endpoint events should be captured for security monitoring. Since the project focuses on threat detection and incident analysis, the selected telemetry categories include process execution, network connection, file creation, registry modification, DNS query, and authentication-related events.

Process execution logs are important because many attacks begin with suspicious command execution or script usage. Network connection logs help identify unusual outbound communication. File creation events can indicate the creation of scripts, payloads, or suspicious files. Registry modification events may indicate persistence mechanisms. DNS query logs can support investigation of suspicious domain communication.

The telemetry collection strategy was designed to provide enough visibility for detection while avoiding unnecessary log noise. This supports the goal of reducing irrelevant information and focusing on security-relevant events [14] [15].

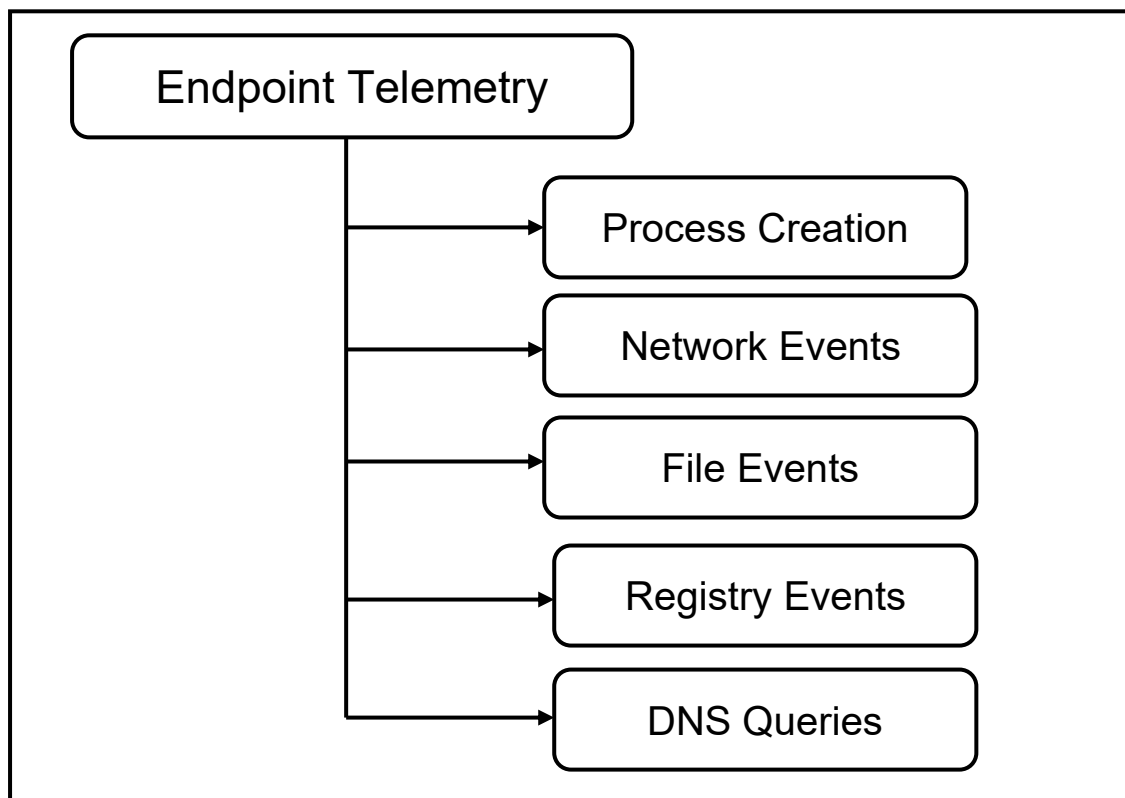


Figure 3.2: Endpoint Telemetry Categories Selected for Monitoring.

Telemetry Category	Security Relevance
Process Creation	Detect suspicious command execution
Network Connection	Identify unusual communication
File Creation	Observe creation of scripts or payloads
Registry Modification	Detect possible persistence activity
DNS Query	Investigate suspicious domain access
Login Events	Identify authentication-related activity

Table 3.4: Telemetry Categories and Security Relevance.

3.6 Data Flow Methodology.

The data flow methodology explains how endpoint telemetry moves from the source endpoint to the SIEM platform. The flow begins with user or system activity on the Windows virtual machine. Sysmon captures relevant events and stores them in Windows Event Logs. Azure Monitor Agent collects the selected logs based on Data Collection Rule configuration and forwards them to Log Analytics Workspace. Microsoft Sentinel then uses the collected telemetry for detection, alerting, and investigation.

This data flow was designed to demonstrate an end-to-end monitoring pipeline similar to a SOC environment. This approach allows security analysts to view endpoint data from a centralized platform instead of manually checking local endpoint logs [14] [15].

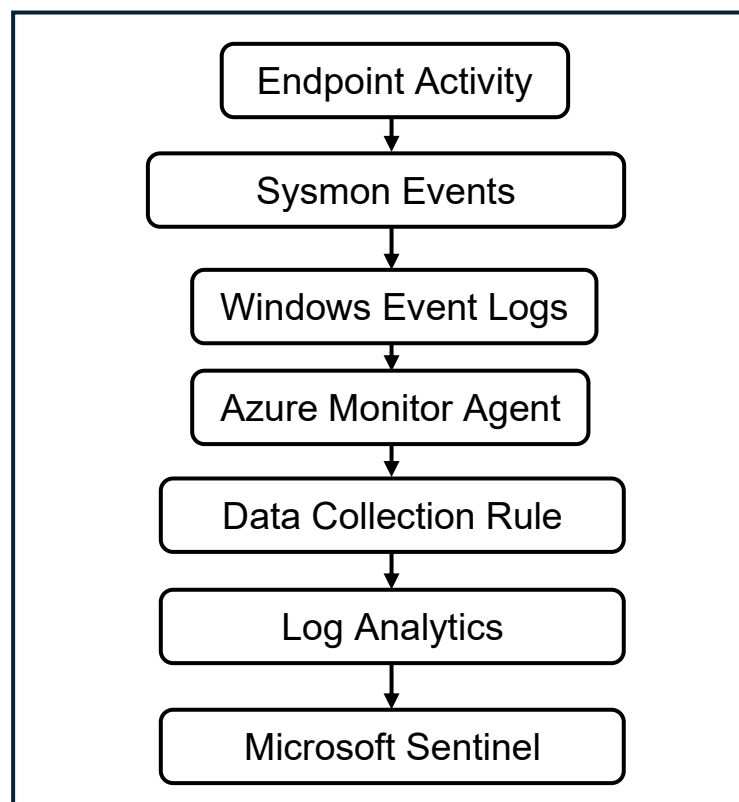


Figure 3.3: Designed Telemetry Data Flow.

3.7 Detection Methodology.

The detection methodology was designed using rule-based analysis. In this approach, suspicious activities are identified by applying predefined query logic to endpoint telemetry. Kusto Query Language was selected for querying logs in Log Analytics Workspace and Microsoft Sentinel.

The detection logic was planned around common endpoint security scenarios such as suspicious PowerShell execution, abnormal process creation, repeated failed login attempts, network activity from unusual processes, and creation of new local users. These scenarios were selected because they represent activities frequently investigated in SOC environments.

The detection methodology does not claim to identify every possible cyberattack. Instead, it demonstrates how endpoint telemetry can be used to detect selected suspicious behaviors and support security investigation [9] [16].

Detection Scenario	Telemetry Source	Purpose
PowerShell execution	Process logs	Identify script-based activity
Suspicious command-line usage	Process logs	Detect unusual command execution
Repeated failed logins	Security logs	Detect possible brute-force attempts
New local user creation	Security logs	Identify possible persistence activity
Network connection from process	Sysmon network events	Analyze outbound communication
Registry modification	Sysmon registry events	Detect possible persistence changes

Table 3.5: Planned Detection Scenarios.

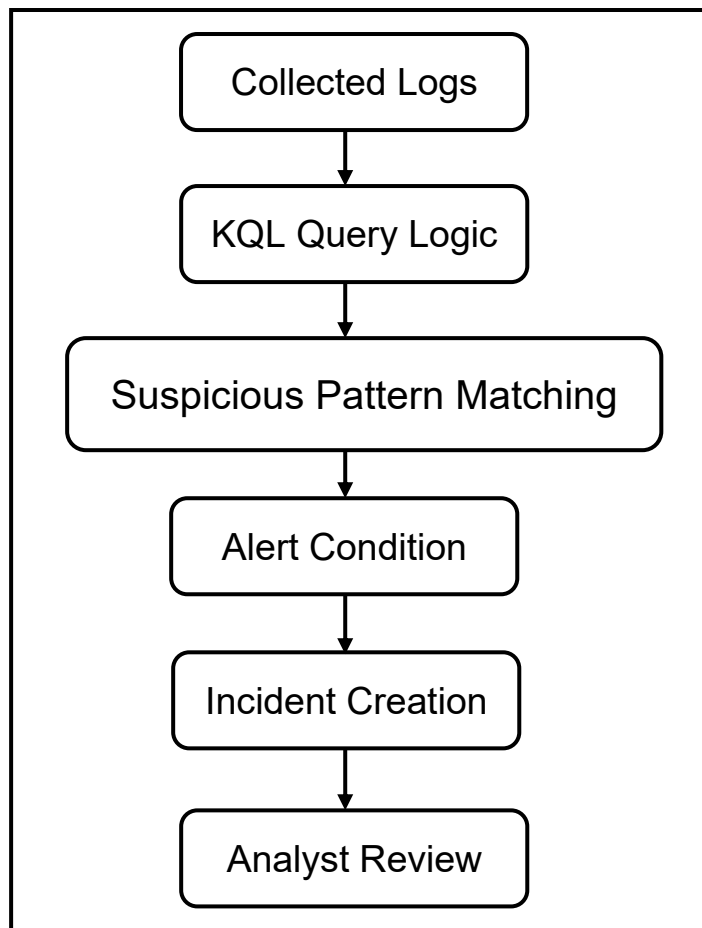


Figure 3.4: Detection Methodology Workflow.

3.8 Incident Analysis Methodology.

The incident analysis methodology defines how generated alerts should be reviewed and interpreted. Once a detection rule identifies suspicious activity, the alert is analyzed by reviewing the related event details, timestamp, affected host, command line, user account, and associated process information.

The investigation approach follows a structured sequence. First, the alert is validated to determine whether the activity is expected or suspicious. Second, related telemetry is reviewed to understand the context of the event. Third, the activity is mapped to a possible attack behavior, such as execution, persistence, credential access, or discovery. Finally, the incident is documented with observations and severity.

This methodology helps convert raw endpoint logs into meaningful security findings. It also supports the role of a SOC analyst by providing a repeatable process for investigating alerts.

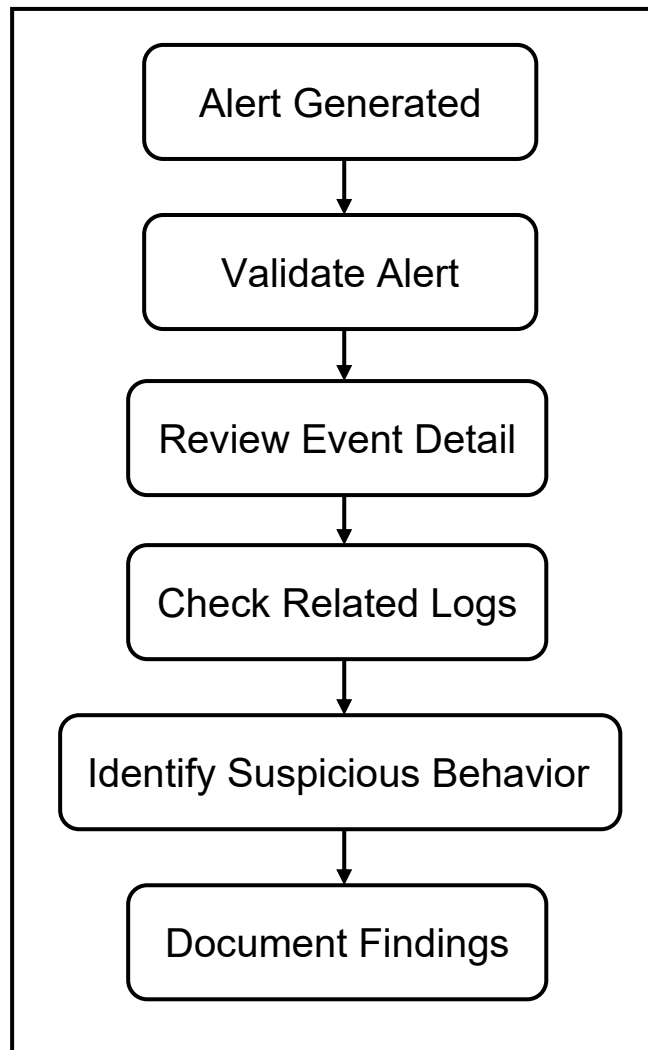


Figure 3.5 Incident Analysis Workflow.

3.9 Ethical and Practical Scope of Implementation.

The implementation was designed within a controlled lab environment to avoid risk to any production or client systems. The project uses simulated endpoint activity and lab-generated telemetry only. No confidential organizational data, client data, or production security logs are used.

This controlled approach ensures that the research remains ethical, reproducible, and safe. It also allows testing of detection logic without impacting real users or enterprise infrastructure. The scope of the project is limited to demonstrating telemetry collection, SIEM integration, basic threat detection, and incident analysis in a lab environment.

3.10 Chapter Summary.

This chapter presented the design and methodology of the proposed endpoint telemetry-based threat detection system. The system was designed using a layered architecture consisting of endpoint, telemetry generation, collection, storage, detection, and analysis layers. The methodology focused on controlled lab-based implementation, selected telemetry collection, KQL-based detection logic, and structured incident analysis.

Chapter 4

Implementation and Results Analysis

4.1 Introduction.

This chapter presents the practical implementation and results analysis of the proposed real-time threat detection and incident analysis system using endpoint telemetry. The methodology and system design were discussed in Chapter 3, while this chapter focuses on the actual deployment, configuration, telemetry collection, query execution, alert generation, incident investigation, and result interpretation.

The implementation was carried out in a controlled lab environment using a Windows virtual machine as the endpoint system. Endpoint telemetry was generated through Sysmon and Windows Event Logs, collected using Azure Monitor Agent and Data Collection Rules, stored in Log Analytics Workspace, and analyzed using Microsoft Sentinel. The implementation demonstrates how endpoint-level security events can be converted into meaningful alerts and incidents for security analysis.

The purpose of this chapter is not only to show that the tools were configured but also to evaluate whether the designed monitoring pipeline successfully achieved the dissertation objectives. The results are analyzed using screenshots, KQL query outputs, generated alerts, incident evidence, tables, and graphs.

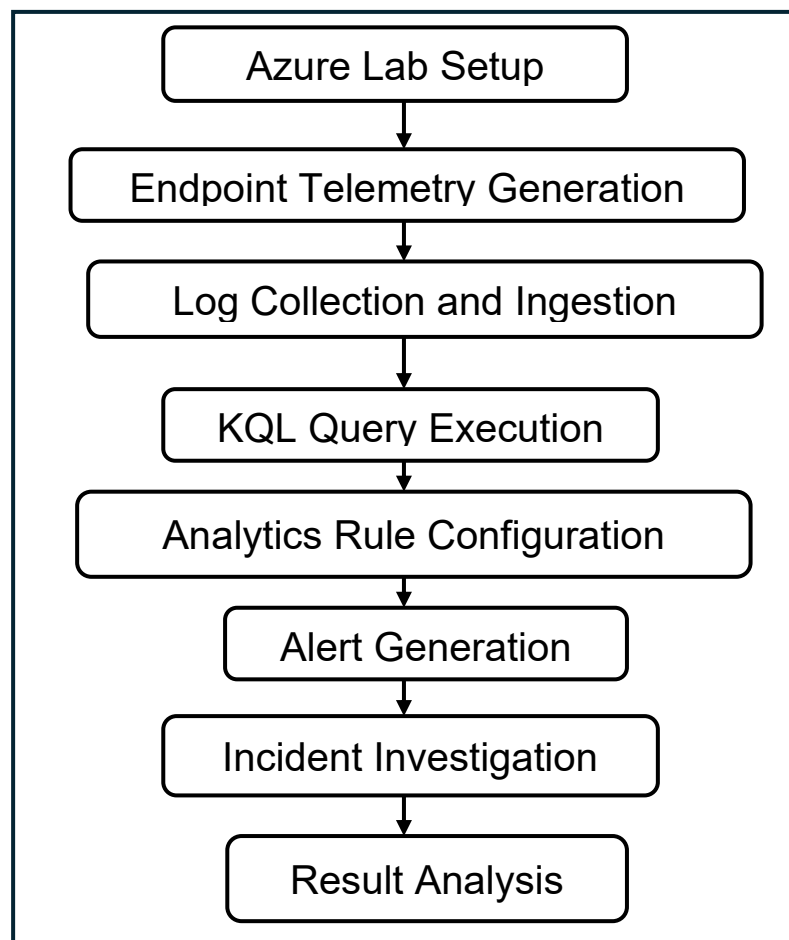


Figure 4.1: Implementation and Validation Flow.

4.2 Azure Lab Environment Implementation.

The first phase of implementation involved creating the required Azure lab environment. A dedicated resource group was created to organize all resources used for the dissertation project. A Windows virtual machine was deployed inside this environment to act as the monitored endpoint. This virtual machine was used to generate endpoint activity such as process execution, login events, command-line activity, and system-level events.

The Azure environment was selected because it supports cloud-based monitoring, centralized log collection, and integration with Microsoft Sentinel. The use of a cloud-based lab also made the project easier to manage, reproduce, and document. Since the implementation was performed in a controlled environment, no production or client data was used.

This phase confirmed that the basic infrastructure required for endpoint telemetry collection and security monitoring was successfully created.

Component	Purpose	Status
Resource Group	Organized project resources	Implemented
Windows Virtual Machine	Acted as monitored endpoint	Implemented
Log Analytics Workspace	Stored collected telemetry	Implemented
Azure Monitor Agent	Forwarded endpoint logs	Implemented
Data Collection Rule	Defined log collection scope	Implemented
Microsoft Sentinel	Provided SIEM and investigation capabilities	Implemented

Table 4.1: Azure Lab Components Implemented.

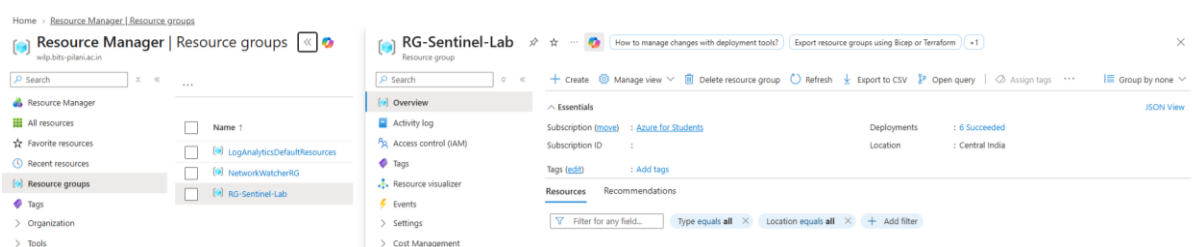


Figure 4.2: Resource Group Created for Dissertation Lab.

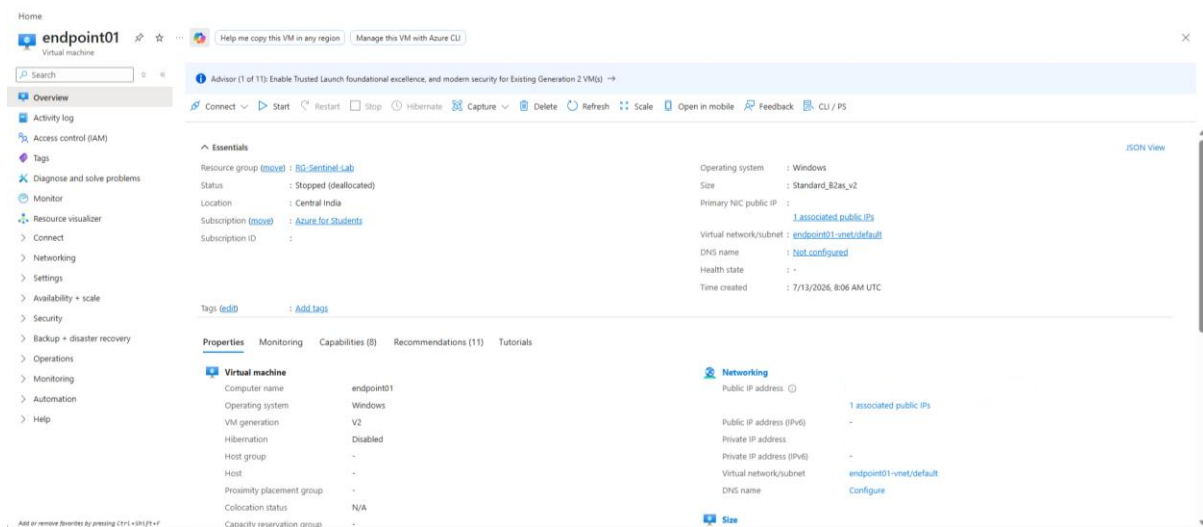


Figure 4.3: Windows Virtual Machine Used as Monitored Endpoint.

Figure 4.3 shows the Windows virtual machine deployed for the dissertation lab. This VM acted as the endpoint system from which telemetry was generated and forwarded to the monitoring platform.

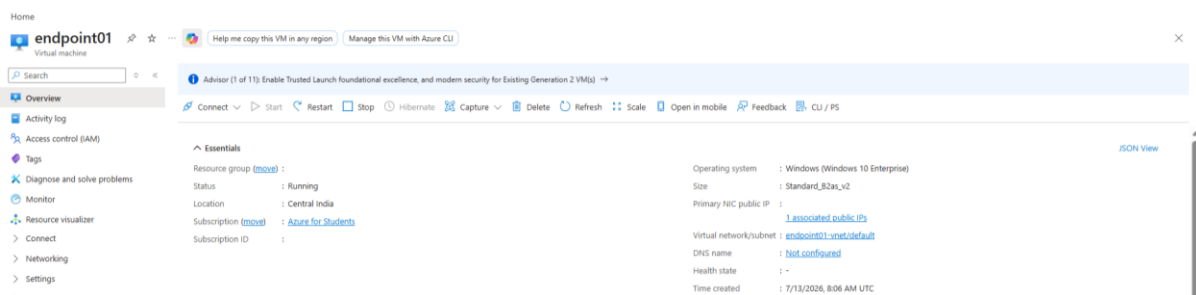


Figure 4.4: Operational Status of Endpoint Virtual Machine.

4.3 Sysmon Deployment and Endpoint Telemetry Generation.

Sysmon was deployed on the Windows virtual machine to collect detailed endpoint telemetry. Sysmon improves endpoint visibility by recording important events such as process creation, network connections, file activity, registry modifications, and DNS queries. These events are highly useful for detecting suspicious behavior because many cyberattacks involve abnormal process execution, script usage, file creation, persistence changes, or network communication.

After installation, the Sysmon service was verified to ensure that it was running properly. The generated logs were reviewed in Windows Event Viewer under the Sysmon Operational log. This confirmed that the endpoint was successfully generating telemetry required for threat detection and incident analysis.

Sysmon is described by Microsoft as a Windows system service and device driver that monitors and logs system activity to the Windows Event Log, including details about process creation, network connections, and file creation time changes. Microsoft also states that Sysmon events can help detect malicious or unusual behavior when collected through Windows Event Collection or a SIEM agent [12] [18].

Event ID	Event Type	Security Importance
1	Process Creation	Helps detect suspicious command execution
3	Network Connection	Helps identify external or unusual communication
11	File Creation	Helps observe creation of scripts or suspicious files
13	Registry Value Set	Helps identify possible persistence activity
22	DNS Query	Helps investigate domain lookups and network behavior

Table 4.2: Sysmon Events Considered for Security Monitoring.

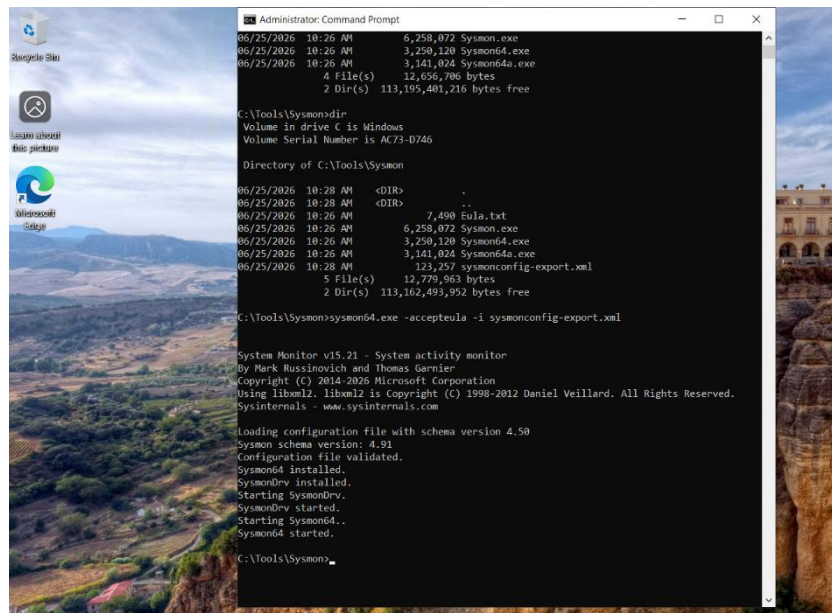


Figure 4.5: Sysmon Installation on Windows Endpoint.

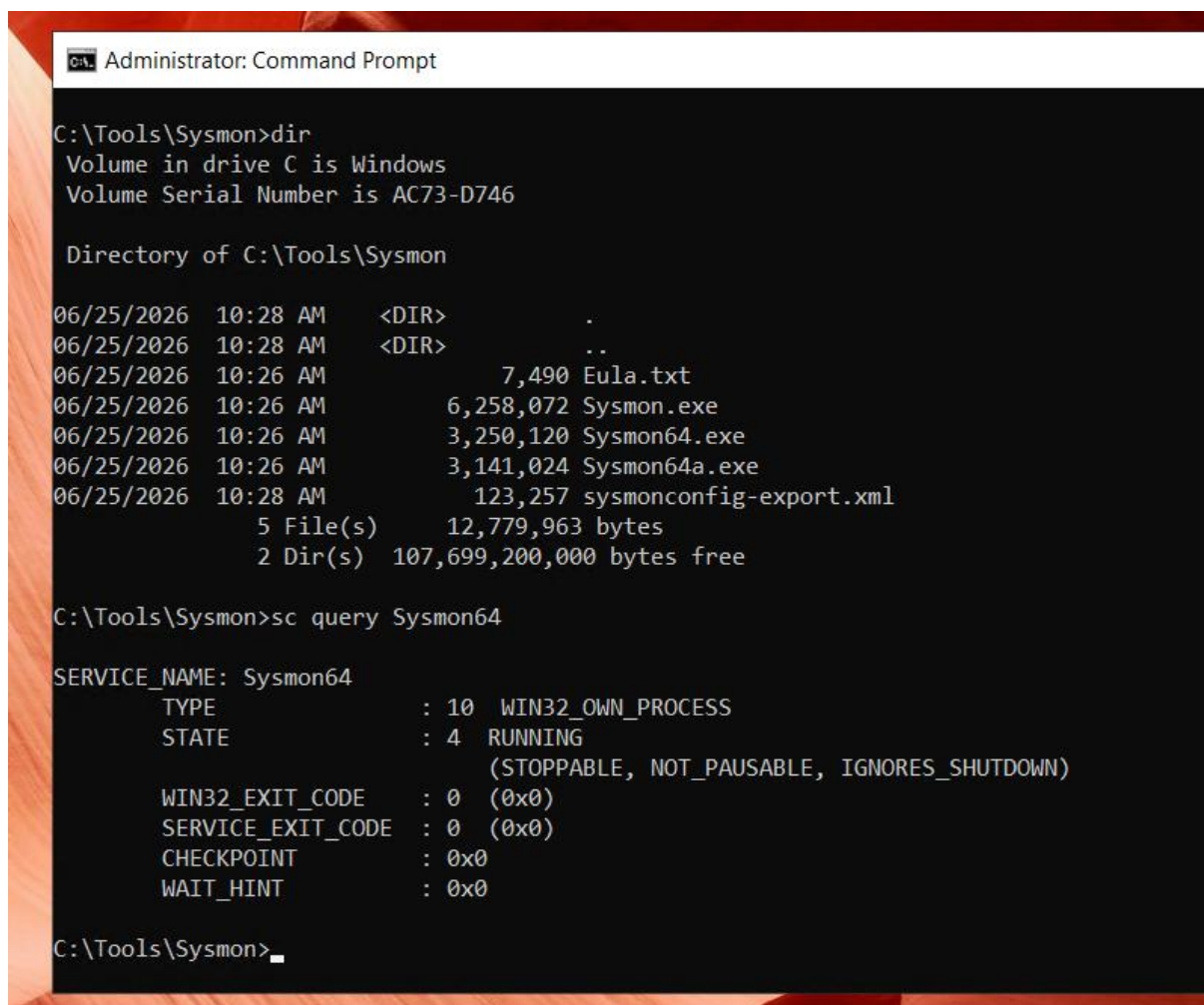


Figure 4.6: Verification of Sysmon Service Status.

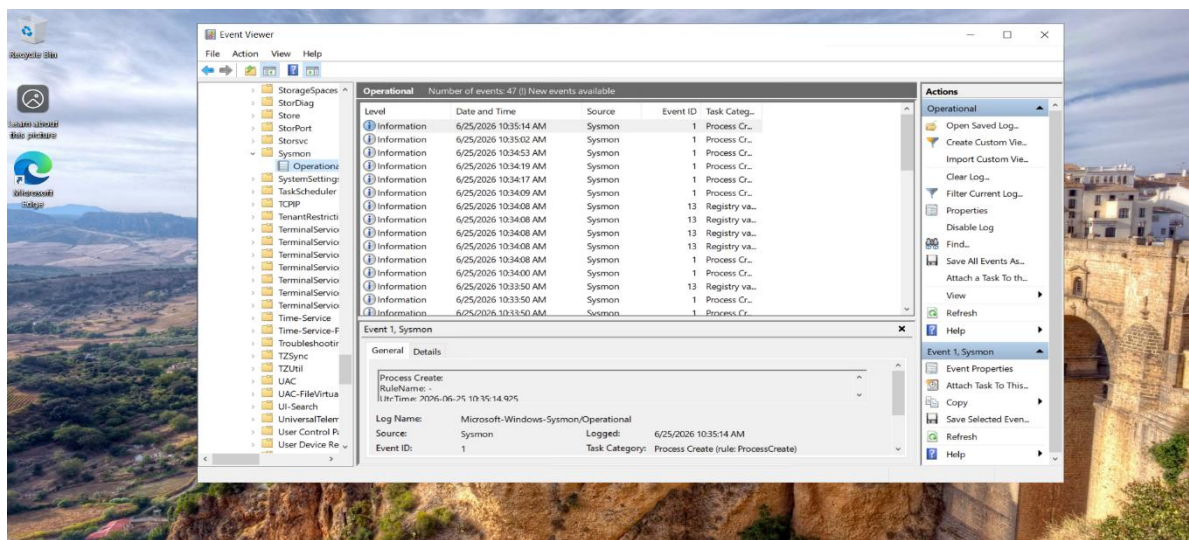


Figure 4.7: Sysmon Events Generated on Endpoint.

The Sysmon Operational log confirmed that endpoint telemetry was being generated successfully. This telemetry later served as the input for centralized log collection and detection rule development.

4.4 Log Collection and Telemetry Ingestion into Azure.

After endpoint telemetry was generated locally, the next phase involved forwarding the collected logs to Azure. Azure Monitor Agent was configured on the Windows virtual machine to collect selected guest operating system logs and send them to Log Analytics Workspace. Data Collection Rules were used to define what data should be collected and where it should be sent.

This phase was important because local endpoint logs are difficult to analyze at scale. By forwarding logs to Log Analytics Workspace, the telemetry became centrally available for querying, analysis, and SIEM-based detection. The successful ingestion of logs was validated by running KQL queries in Log Analytics Workspace.

Azure Monitor Agent collects monitoring data from the guest operating system of Azure and hybrid virtual machines and delivers the data to Azure Monitor for use by services such as Microsoft Sentinel. Azure Monitor Agent collects data according to Data Collection Rules, and those rules define what data is collected, processed, and sent [10] [14] [15].

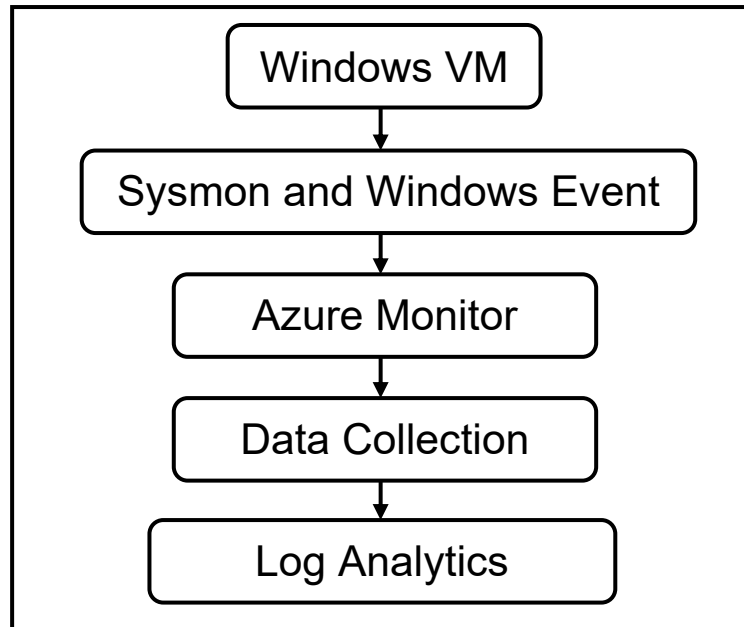


Figure 4.8: Actual Telemetry Ingestion Path.

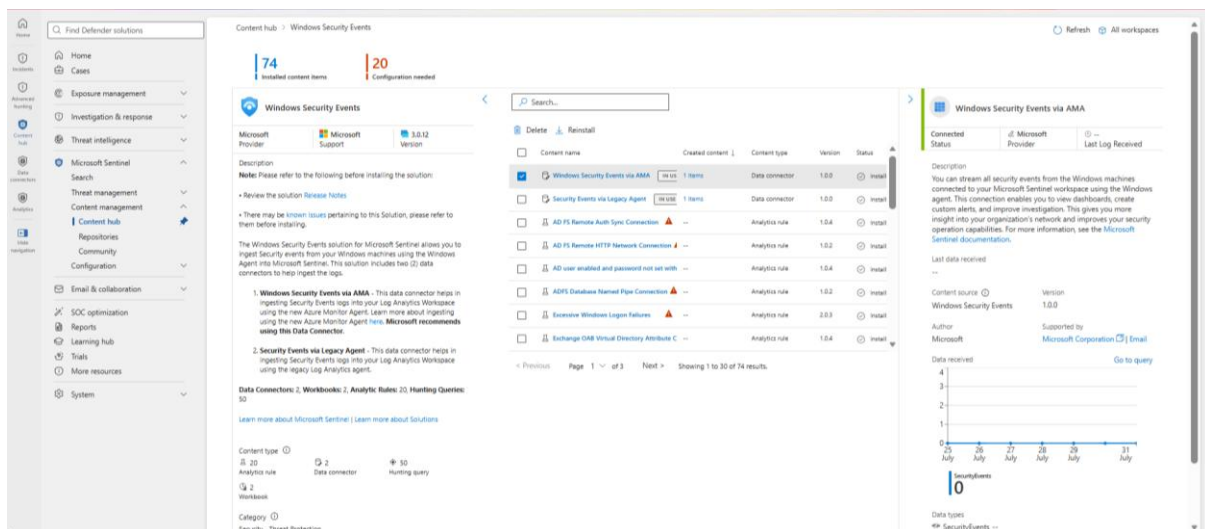


Figure 4.9: Azure Monitor Agent Configuration & Installation.

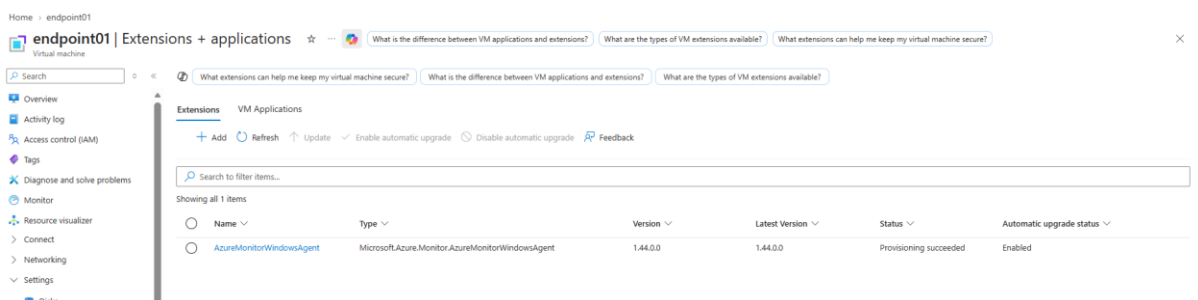


Figure 4.10: Azure Monitor Agent Installed on Virtual Machine.

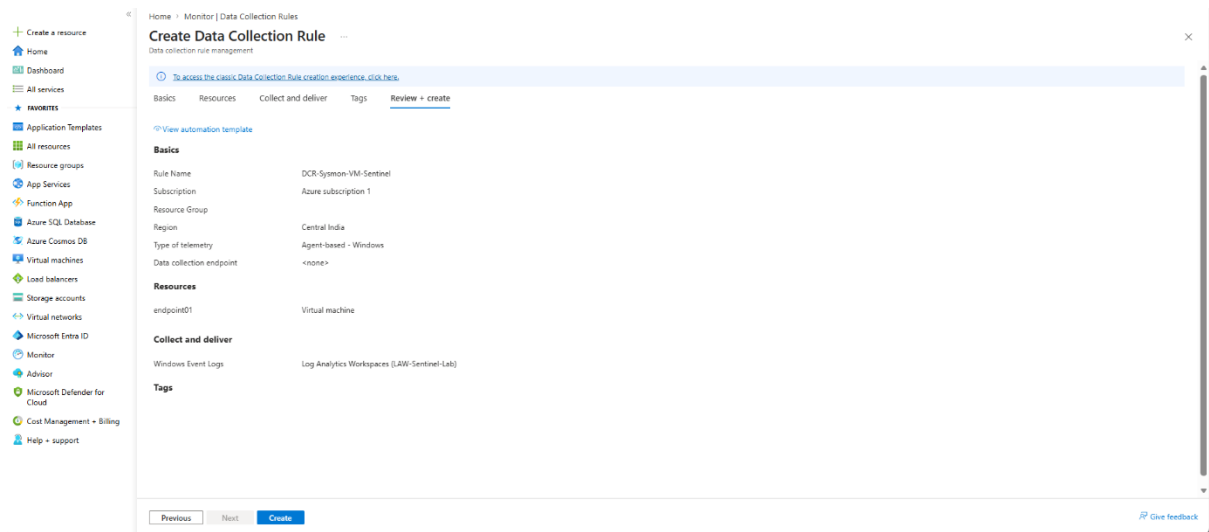


Figure 4.11: Data Collection Rule Used for Endpoint Log Collection.

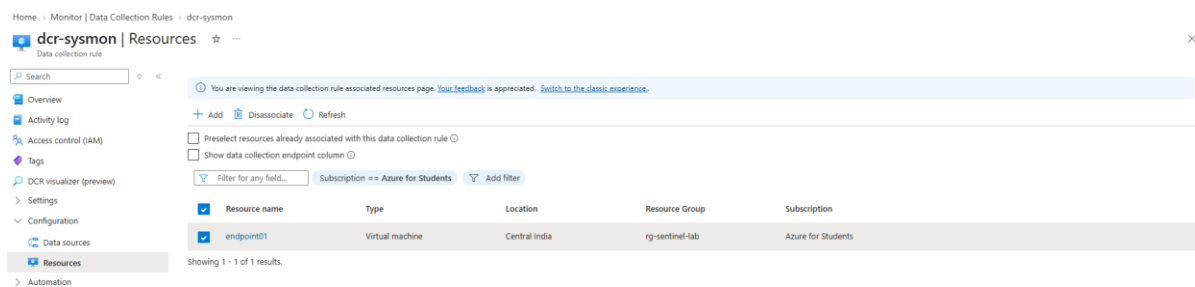


Figure 4.12: Association of Data Collection Rule with Endpoint VM.

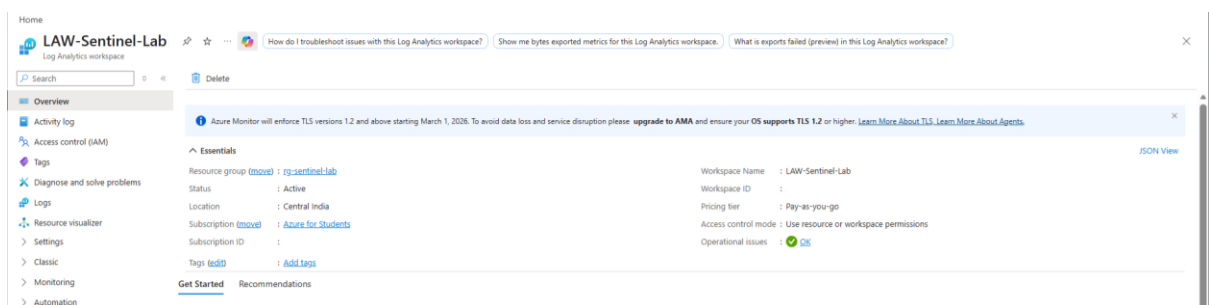


Figure 4.13: Log Analytics Workspace Configuration.

TimeGenerated (UTC)	Account	AccountType	Computer	EventSourceName	Channel	Task	Level	EventData	EventID
8/5/2026 9:49:25:053 AM	endpoint01\corpuser01	User	endpoint01	Microsoft-Windows-Security-Auditing	Security	13824	0	<EventData xmlns="http://schemas.microsoft.com/win/200...	4798
8/5/2026 9:49:16:008 AM	NT AUTHORITY\SYSTEM	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	12548	0		4672
8/5/2026 9:49:16:007 AM	NT AUTHORITY\SYSTEM	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	12544	0		4624
8/5/2026 9:49:13:946 AM	NT AUTHORITY\SYSTEM	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	12548	0		4672
8/5/2026 9:49:13:946 AM	NT AUTHORITY\SYSTEM	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	12544	0		4624
8/5/2026 9:49:13:921 AM	NT AUTHORITY\SYSTEM	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	12548	0		4672
8/5/2026 9:49:13:921 AM	NT AUTHORITY\SYSTEM	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	12544	0		4624
8/5/2026 9:44:49:124 AM	NT AUTHORITY\SYSTEM	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	12548	0		4672
8/5/2026 9:44:49:124 AM	NT AUTHORITY\SYSTEM	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	12544	0		4624
8/5/2026 8:14:26:458 AM	endpoint01	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	13824	0	<EventData xmlns="http://schemas.microsoft.com/win/200...	5379
8/5/2026 8:14:26:457 AM	endpoint01	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	13824	0	<EventData xmlns="http://schemas.microsoft.com/win/200...	5379
8/5/2026 8:14:26:455 AM	endpoint01	Machine	endpoint01	Microsoft-Windows-Security-Auditing	Security	13824	0	<EventData xmlns="http://schemas.microsoft.com/win/200...	5379

Figure 4.14: Endpoint Logs Available in Log Analytics Workspace.

Depending on the connector and data collection configuration, Sysmon telemetry may appear under different tables. In this implementation, the available log table was validated using Log Analytics queries before building detection logic.

4.5 Microsoft Sentinel Configuration.

Microsoft Sentinel was enabled on the Log Analytics Workspace to provide SIEM-based detection, alerting, and incident investigation capabilities. After enabling Sentinel, the collected endpoint telemetry became available for security monitoring and detection rule development.

Microsoft Sentinel was used to create analytics rules, generate alerts, review incidents, and support structured investigation. This implementation provided a SOC-style workflow where raw endpoint telemetry was converted into security alerts and incidents.

Microsoft Sentinel is described by Microsoft Learn as an AI-first, cloud-native SIEM platform that consolidates and analyzes security data at scale and provides tools for detecting, investigating, and responding to threats across hybrid and multicloud environments. Microsoft documentation also states that Microsoft Sentinel provides attack detection, threat visibility, proactive hunting, and threat response [10] [13].

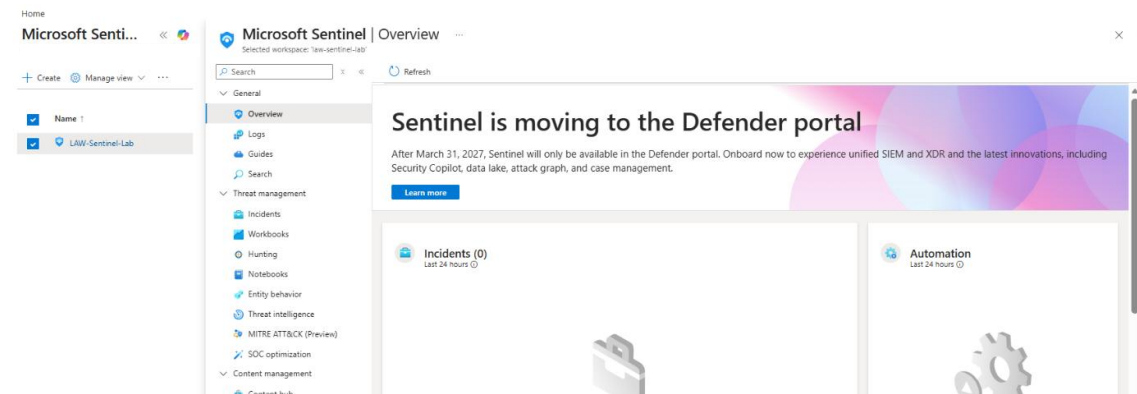


Figure 4.15: Microsoft Sentinel Enabled on Log Analytics Workspace.

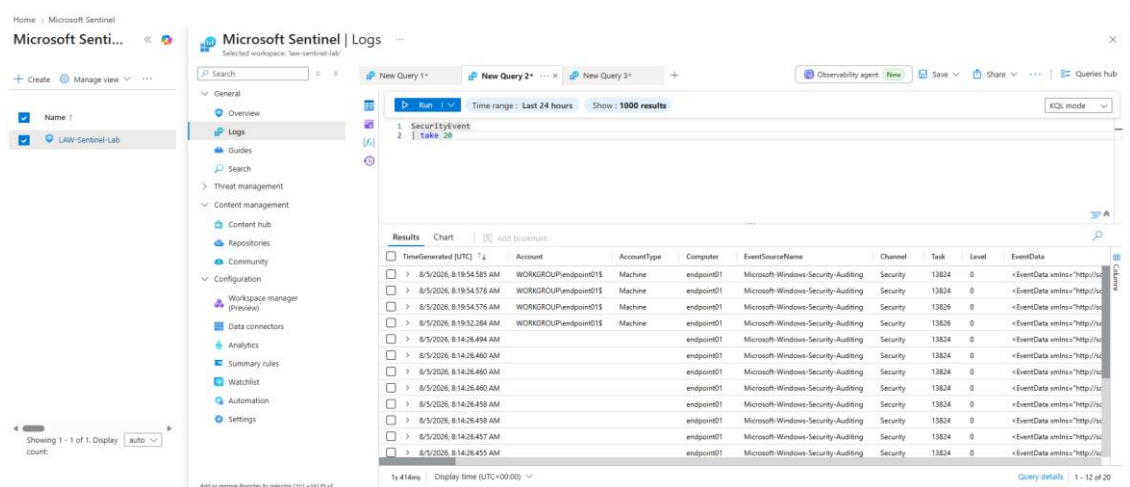


Figure 4.16: Log Query Interface in Microsoft Sentinel.

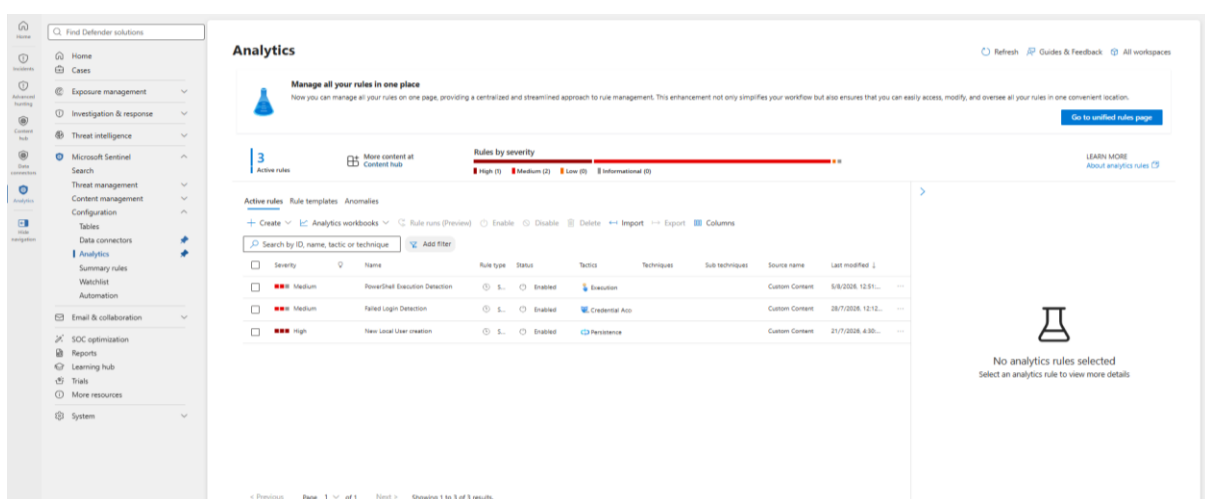


Figure 4.17: Analytics Rule Management in Microsoft Sentinel.

This configuration confirmed that the telemetry collection layer and SIEM analysis layer were successfully integrated.

4.6 KQL-Based Detection Query Execution.

Kusto Query Language was used to analyze the collected telemetry in Log Analytics Workspace and Microsoft Sentinel. KQL queries helped filter endpoint logs based on event type, activity description, process execution, authentication behavior, and suspicious command-line patterns.

The objective of this phase was to verify that collected telemetry could be searched and analyzed for detection purposes. Queries were executed to identify process activity, PowerShell usage, failed login events, and Sysmon-generated endpoint events. The results confirmed that endpoint telemetry was available for investigation and could support detection logic.

Query 1: Basic Event Validation:

```
Event
| take 20
```

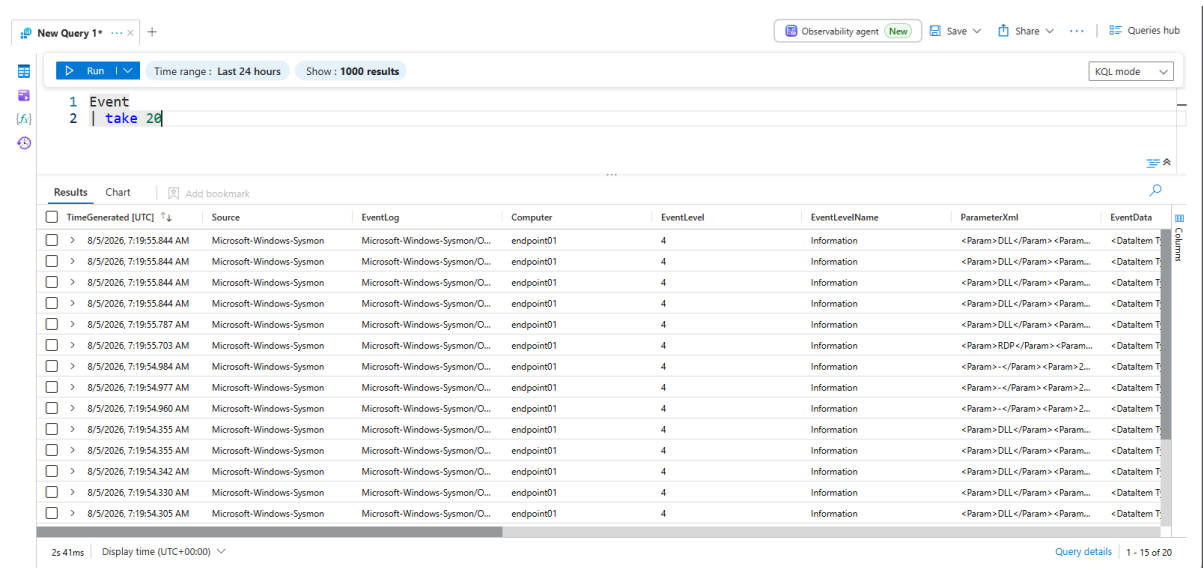


Figure 4.18: Basic Log Validation Query Output.

Query 2: Sysmon Events:

Event
| where Source contains "Sysmon"
| project TimeGenerated, Computer, Source, EventID, RenderedDescription
| take 20

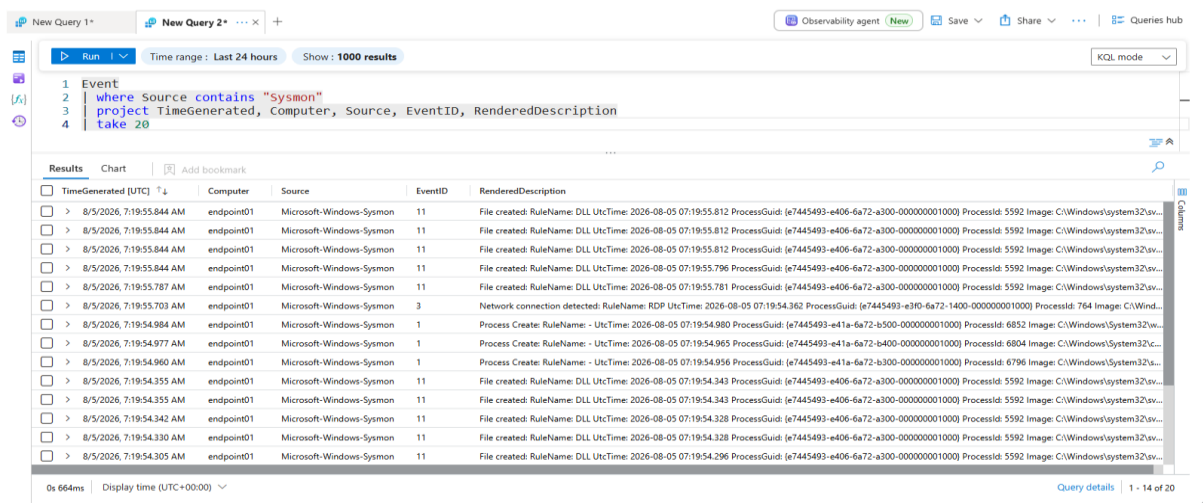


Figure 4.19: Sysmon Event Query Results.

Query 3: PowerShell Activity:

Event
| where RenderedDescription contains "powershell"
| project TimeGenerated, Computer, EventID, RenderedDescription

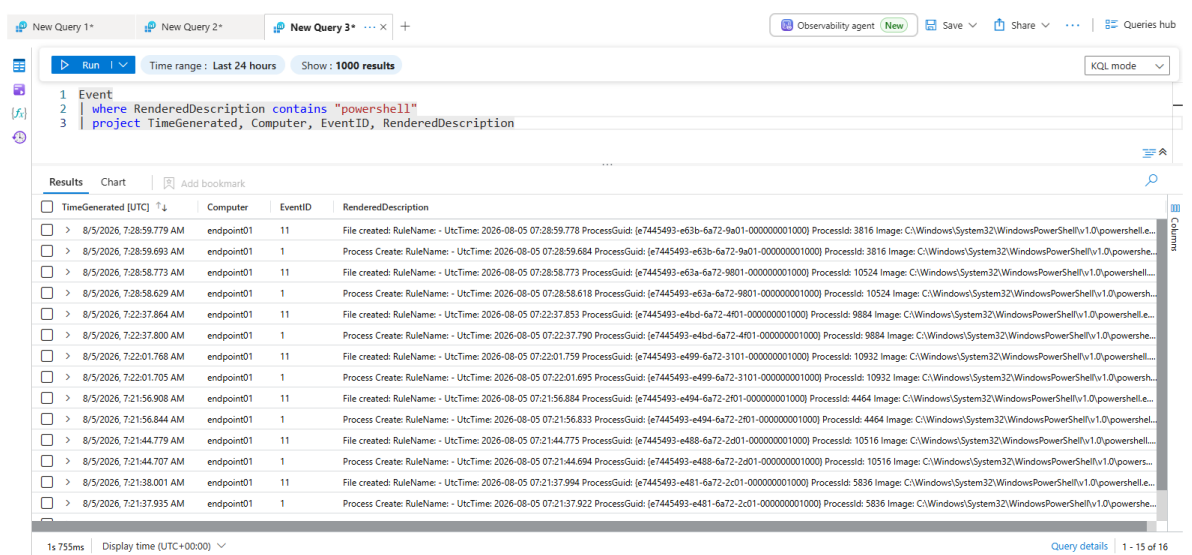


Figure 4.20: PowerShell Activity Detection Using KQL.

Query 4: Failed Login Events: SecurityEvent

| where EventID == 4625

| project TimeGenerated, Computer, Account, Activity

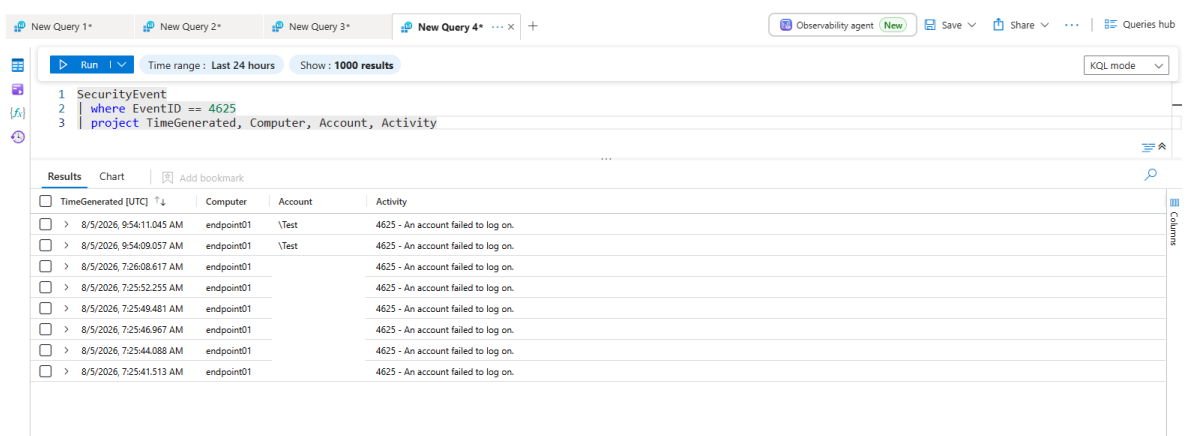


Figure 4.21: Failed Authentication Event Analysis.

Query 5: Process Creation Events:

SecurityEvent

| where EventID == 4688

| project TimeGenerated, Computer, Account, Process, CommandLine

TimeGenerated [UTC]	Computer	Account	Process	CommandLine
> 8/5/2026, 7:19:12.170 AM	endpoint01	SYSTEM	lsass.exe	
> 8/5/2026, 7:19:12.099 AM	endpoint01	SYSTEM	services.exe	
> 8/5/2026, 7:19:12.091 AM	endpoint01	SYSTEM	winlogon.exe	
> 8/5/2026, 7:19:12.057 AM	endpoint01	SYSTEM	csrss.exe	
> 8/5/2026, 7:19:12.050 AM	endpoint01	SYSTEM	wininit.exe	
> 8/5/2026, 7:19:12.048 AM	endpoint01	SYSTEM	smss.exe	
> 8/5/2026, 7:19:11.972 AM	endpoint01	SYSTEM	csrss.exe	
> 8/5/2026, 7:19:11.801 AM	endpoint01	SYSTEM	smss.exe	
> 8/5/2026, 7:19:09.496 AM	endpoint01	SYSTEM	autochk.exe	
> 8/5/2026, 7:18:52.279 AM	endpoint01	SYSTEM	smss.exe	
> 8/5/2026, 7:18:52.278 AM	endpoint01	SYSTEM	Registry	

Figure 4.22: Process Creation Event Analysis.

The query result proves that the endpoint telemetry was successfully collected and could be filtered for security investigation.

4.7 Detection Use Case Implementation.

The detection use cases were implemented to demonstrate how endpoint telemetry can identify security-relevant behavior. The selected use cases were based on common SOC investigation scenarios and were suitable for a controlled lab environment.

The implemented detection use cases included PowerShell activity detection, failed login monitoring, process execution monitoring, and Sysmon event monitoring. These use cases were selected because they represent activities commonly reviewed during endpoint security investigations.

The goal was not to simulate a real attack but to demonstrate how telemetry-driven monitoring can detect selected suspicious behaviors and support analyst investigation.

Use Case	Detection Purpose	Log Source	Expected Output
PowerShell Activity Detection	Identify script execution activity	Sysmon or Windows Event Logs	Query result or alert
Failed Login Detection	Identify unsuccessful login attempts	SecurityEvent	Authentication event records
Process Creation Monitoring	Observe process execution	SecurityEvent or Sysmon	Process activity details
Sysmon Event Monitoring	Validate endpoint telemetry	Sysmon Operational Logs	Sysmon event output

Table 4.3: Detection Use Cases Implemented.

Analytics > Analytics > Analytics rule wizard

Validation passed.

Analytics rule wizard - Edit existing Scheduled rule

PowerShell Execution Detection

- General
- Set rule logic
- Incident settings
- Automated response
- Review + create

Analytics rule details

Name: PowerShell Execution Detection

Description: Detects PowerShell process execution on endpoint systems.

MITRE ATT&CK: Execution

Severity: Medium

Status: Enabled

Analytics rule settings

Rule query: SecurityEvent | where EventID == 4688 | project TimeGenerated, Computer, Account, Process, CommandLine

Rule frequency: Run query every 5 minutes

Rule period: Last 10 minutes data

Rule start time: Automatic

Rule threshold: Trigger alert if query returns more than 0 results

Event grouping: Trigger an alert for each event

Suppression: Not configured

Entity mapping: Not configured

Custom details: Not configured

Alert details: Not configured

Incident settings: Create incidents from this rule: Enabled

< Previous Save Cancel

Figure 4.23: Analytics Rule for PowerShell Execution Detection.

Analytics > Analytics > Analytics rule wizard

Validation passed.

Analytics rule wizard - Edit existing Scheduled rule

Failed Login Detection

- General
- Set rule logic
- Incident settings
- Automated response
- Review + create

Analytics rule details	Failed Login Detection
Name	Failed Login Detection
Description	Detects multiple failed login attempts on the endpoint machine.
MITRE ATT&CK	Credential Access
Severity	Medium
Status	Enabled
Analytics rule settings	
Rule query	SecurityEvent where EventID == 4625 project TimeGenerated, Computer, Account, Activity
Rule frequency	Run query every 5 minutes
Rule period	Last 10 minutes data
Rule start time	Automatic
Rule threshold	Trigger alert if query returns more than 0 results
Event grouping	Trigger an alert for each event
Suppression	Not configured
Entity mapping	Not configured
Custom details	Not configured
Alert details	Not configured
Incident settings	
Create incidents from this rule	Enabled

< Previous Save Cancel

Figure 4.24: Analytics Rule for Failed Login Detection.

Analytics > Analytics > Analytics rule wizard

Validation passed.

Analytics rule wizard - Edit existing Scheduled rule

New Local User creation

- General
- Set rule logic
- Incident settings
- Automated response
- Review + create

Analytics rule details	New Local User creation
Name	New Local User creation
Description	Detects creation of a new local account.
MITRE ATT&CK	Persistence
Severity	High
Status	Enabled
Analytics rule settings	
Rule query	SecurityEvent where EventID == 4720 project TimeGenerated, Computer, SubjectAccount, TargetAccount, Activity
Rule frequency	Run query every 5 minutes
Rule period	Last 30 minutes data
Rule start time	Automatic
Rule threshold	Trigger alert if query returns more than 0 results
Event grouping	Trigger an alert for each event
Suppression	Not configured
Entity mapping	Not configured
Custom details	Not configured
Alert details	Not configured
Incident settings	
Create incidents from this rule	Enabled

< Previous Save Cancel

Figure 4.25: Analytics Rule for Local User Creation Detection.

The screenshot shows the 'Incidents' page in Microsoft Sentinel. It displays a table of recent incidents and alerts. The table has columns for Incident name, Incident ID, Priority score, Severity, Investigation state, Categories, Impacted assets, Active alerts, Service sources, and Detection source. Three incidents are listed: 'New Local User creation' (High severity, Persistence category), 'PowerShell Execution Detection' (Medium severity, Execution category), and 'Failed Login Detection' (Medium severity, Credential access category).

Incident name	Incident ID	Priority score	Severity	Investigation state	Categories	Impacted assets	Active alerts	Service sources	Detection source
New Local User creation	433408	1	High		Persistence		1/1	Microsoft Sentinel	Scheduled detecti
PowerShell Execution Detection	433402	1	Medium		Execution		1/1	Microsoft Sentinel	Scheduled detecti
Failed Login Detection	433388	1	Medium		Credential access		1/1	Microsoft Sentinel	Scheduled detecti

Figure 4.26: Generated Alerts for Analytics Rules.

The detection use cases were tested using lab-generated activities to avoid risk to production systems or real users.

4.8 Analytics Rule Configuration in Microsoft Sentinel.

Analytics rules were created in Microsoft Sentinel to automate the detection of selected suspicious activities. Instead of manually executing queries every time, analytics rules evaluate incoming telemetry based on predefined logic and generate alerts when the defined conditions are met.

The custom analytics rules were based on the detection use cases implemented in the lab. These rules helped convert raw log data into actionable alerts. This phase demonstrated how query-based detection logic can be operationalized in a SIEM environment.

4.8.1 MITRE ATT&CK Mapping of Detection Rules.

The implemented detection rules were aligned with selected MITRE ATT&CK techniques to better understand the security significance of the detected activities. Mapping the detections with MITRE ATT&CK provides a structured approach for relating endpoint telemetry to real-world attacker behaviors.

PowerShell execution detection was associated with the Execution tactic because attackers frequently use PowerShell to execute commands and scripts on compromised systems.

Failed login detection was associated with Credential Access due to its relevance in identifying password guessing and brute-force attempts.

Local user creation detection was mapped to Persistence because attackers may create additional user accounts to maintain access.

Process creation monitoring supports Execution and Discovery activities by providing visibility into command execution and system exploration activities.

Detection Use Case	Tactic	Technique
PowerShell Activity Detection	Execution	PowerShell
Failed Login Detection	Credential Access	Brute Force
Local User Creation Detection	Persistence	Create Account
Process Creation Monitoring	Execution	Command and Scripting Interpreter
Sysmon Event Monitoring	Discovery	System Monitoring

Table 4.4: MITRE ATT&CK Mapping of Detection Rules.

Analytics Rule Name	Purpose	Severity	Incident Creation
PowerShell Activity Detection	Detect PowerShell execution	Medium	Enabled
Failed Login Detection	Detect failed authentication attempts	Medium	Enabled
Local User Creation Detection	New Local User Creation	High	Enabled

Table 4.5: Analytics Rules Configured.

Analytics

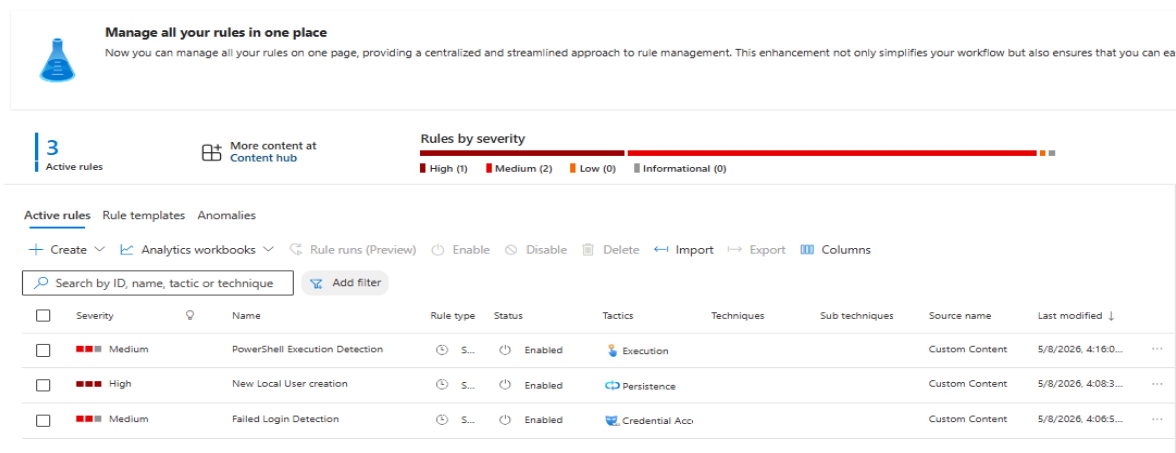


Figure 4.27: Custom Analytics Rules Created in Microsoft Sentinel.

Analytics > Analytics rule wizard

Incident rules: Required information is missing on this rule.

Analytics rule wizard - Create a new Scheduled rule

- General
- Set rule logic
- Incident settings
- Automated response
- Review + create

Create an analytics rule that will run on your data to detect threats.

Analytics rule details

Name *

Description

Severity

MITRE ATT&CK

Status
☒ Enabled

[Next: Set rule logic >](#) [Cancel](#)

Figure 4.28: Analytics Rule General Configuration.

Analytics > Analytics rule wizard

Analytics rule wizard - Create a new Scheduled rule

- General
- Set rule logic
- Incident settings
- Automated response
- Review + create

Define the logic for your new analytics rule.

Rule query
 Any time queries set here will be within the scope defined below in the Query scheduling fields.

Security event

Alert enhancement
☐ Entity mapping
☐ Custom details
☐ Alert details

Query scheduling

Run query every *

Lookback data from the last *

Real time query
☐ Automatically
☒ At specific time (Preferred)

[Previous](#) [Next: Incident settings >](#) [Cancel](#)

Results simulation
 This chart shows the results of the last 10 evaluations of the defined analytics rule. Click a point on the chart to display the raw alerts for that point in time.
[View with current data](#)

Setting a new analytics rule configuration and click 'Test with current data' to test your rule with current data in your workspace.

Figure 4.29: KQL Detection Logic Used in Analytics Rule.

Analytics > Analytics rule wizard

Analytics rule wizard - Create a new Scheduled rule

- General
- Set rule logic
- Incident settings
- Automated response
- Review + create

Incident settings
 Alerts can be grouped together into an incident that should be looked into. You can set whether the alerts that are triggered by this analytics rule should generate incidents.
 Create incidents from alerts triggered by this analytics rule
☒ Enabled

Alert grouping
☒ Microsoft Defender correlation activities can link other alerts or merge existing incidents to the generated incident, regardless of the alert grouping settings defined in the analytics rule.
 Set how the alerts that are triggered by this analytics rule, are grouped into incidents. Grouping alerts into incidents provides the context you need to respond and reduces the noise from single alerts.
 Group related alerts, triggered by this analytics rule, into incidents
☒ Disabled

Limit the group to alerts created within the selected time frame *

Group alerts triggered by this analytics rule into a single incident by
☒ Grouping alerts into a single incident if all the entities match (recommended)
☐ Grouping all alerts triggered by this rule into a single incident
☐ Grouping alerts into a single incident if the selected entity types and details match

Select entities

Select details

[Previous](#) [Next: Automated response >](#) [Cancel](#)

Figure 4.30: Incident Creation Enabled for Analytics Rule.

The analytics rule configuration demonstrated that detection logic can be reused continuously without manual query execution. This improves monitoring efficiency and supports a SOC-style operational workflow.

4.9 Alert Generation Results.

After the analytics rules were configured, the system was tested to verify whether alerts were generated from matching telemetry. When a rule condition matched the collected log data, Microsoft Sentinel generated an alert. This alert represented a security-relevant event requiring review.

Alert generation is a key result of the implementation because it proves that the monitoring pipeline was functional from telemetry collection to detection output. Instead of remaining as raw logs, the endpoint data was converted into actionable security alerts.

Alerts

+ Create

↓ Export

📅 1 Week

7 Alerts

🔍 Search for name or ID

⚙️ Customize columns

Filter set:

Status: New, In progress

✕

🔍 Add filter

🔄 Reset all

Alert name	Tags	Severity	Investigation state	Status	Category	Detection source	Product name	Impacted assets	First activity
☐ Failed Login Detection		🔴🔴🔴 Medium		🔵 New	Credential Access	Scheduled detection	Microsoft Sentinel		5 Aug 2026 16:44
☐ Failed Login Detection		🔴🔴🔴 Medium		🔵 New	Credential Access	Scheduled detection	Microsoft Sentinel		5 Aug 2026 16:44
☐ Failed Login Detection		🔴🔴🔴 Medium		🔵 New	Credential Access	Scheduled detection	Microsoft Sentinel		5 Aug 2026 16:44
☐ Failed Login Detection		🔴🔴🔴 Medium		🔵 New	Credential Access	Scheduled detection	Microsoft Sentinel		5 Aug 2026 16:44
☐ New Local User creation		🔴🔴🔴 High		🔵 New	Persistence	Scheduled detection	Microsoft Sentinel		5 Aug 2026 16:23
☐ PowerShell Execution Detection		🔴🔴🔴 Medium		🔵 New	Execution	Scheduled detection	Microsoft Sentinel		5 Aug 2026 16:09
☐ Failed Login Detection		🔴🔴🔴 Medium		🔵 New	Credential Access	Scheduled detection	Microsoft Sentinel		5 Aug 2026 15:57

Figure 4.31: Security Alert Generated from Endpoint Telemetry.

Alerts

New Local User creation

Part of incident: New Local User creation

View incident page

What happened

Detects creation of a new local account.

ANALYTICS RULE

Analytics rule details

Rule name: New Local User creation

View rule in Sentinel

Rule description: Detects creation of a new local account.

Related events

Query results

View query

1 of 1 selected

Customize columns

TimeGenerated	Activity	Computer	SubjectAccount	TargetAccount
5 Aug 2026 16:28:27 -4720	A user account w...	endpoint01	endpoint01\corpuser01	endpoint01\TestUser1

New Local User creation

🔴🔴🔴 High

🔵 Unknown

🔵 New

Manage alert

Move alert to another incident

Ask Defender Experts

Alert state

Classification: Not Set

Assigned to: Unassigned

Set Classification

Alert details

Alert ID: endb0c-143c-533ba-4d6e-ab1d-0ae5a3204045

Categories: Persistence

MITRE ATT&CK Techniques: -

Detection source: Scheduled detection

Service source: Microsoft Sentinel

Detection technology: -

Generated on: 5 Aug 2026 16:59:05

First activity: 5 Aug 2026 16:28:27

Last activity: 5 Aug 2026 16:28:27

Workspaces: law-sentinel-lab

Evidence

No related evidence was found

Alert description

Figure 4.32: Alert Details Showing Detection Evidence.

The generated alert confirmed that the configured analytics rule successfully detected matching endpoint activity. This validates the connection between log ingestion, query-based detection, and alert creation.

4.10 Incident Investigation Results.

Generated alerts were reviewed as incidents in Microsoft Sentinel. The incident view provided a structured interface for reviewing related alerts, affected entities, event timing, severity, and supporting evidence. This helped in understanding the context of the detected activity and deciding whether it represented expected lab activity or suspicious behavior.

The incident investigation phase demonstrated how endpoint telemetry supports analyst decision-making. Instead of reviewing isolated logs manually, the analyst can use the incident view to examine related information in a single place.

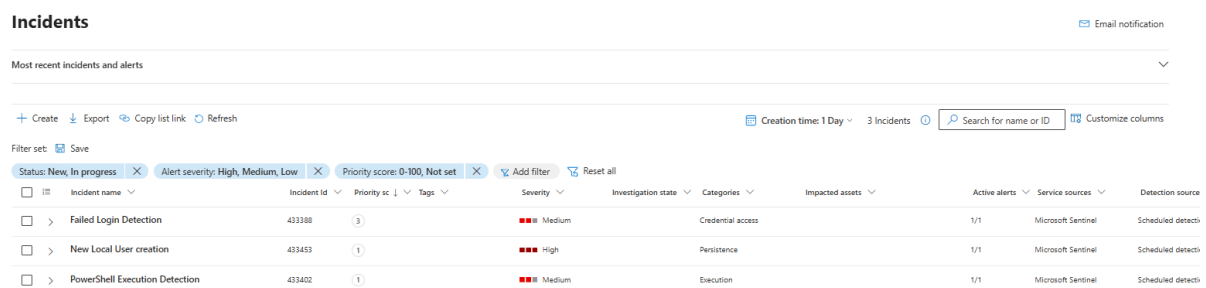
4.10.1 Incident Evidence Summary.

The generated incidents were analyzed to determine whether the configured analytics rules successfully transformed endpoint telemetry into meaningful security events. Each incident was reviewed based on severity, event source, affected host, timestamp, and supporting telemetry evidence.

The results demonstrated that the configured rules successfully detected the intended endpoint activities and produced investigation-ready incidents.

Incident Name	Severity	Source	Result
PowerShell Execution Detection	Medium	Sysmon Logs	Successfully Detected
Failed Login Detection	Medium	SecurityEvent	Successfully Detected
New Local User Creation	High	SecurityEvent	Successfully Detected

Table 4.6: Incident Evidence Summary.



Incidents										Email notification
Most recent incidents and alerts										
+ Create Export Copy list link Refresh										
Creation time: 1 Day 3 Incidents Search for name or ID Customize columns										
Filter set: Save										
Status: New, In progress Alert severity: High, Medium, Low Priority score: 0-100, Not set Add filter Reset all										
☐	Incident name	Incident Id	Priority	Severity	Investigation state	Categories	Impacted assets	Active alerts	Service sources	Detection source
☐	Failed Login Detection	433388	3	Medium		Credential access		1/1	Microsoft Sentinel	Scheduled detects
☐	New Local User creation	433453	1	High		Persistence		1/1	Microsoft Sentinel	Scheduled detects
☐	PowerShell Execution Detection	433402	1	Medium		Execution		1/1	Microsoft Sentinel	Scheduled detects

Figure 4.33: Incident Queue in Microsoft Sentinel.

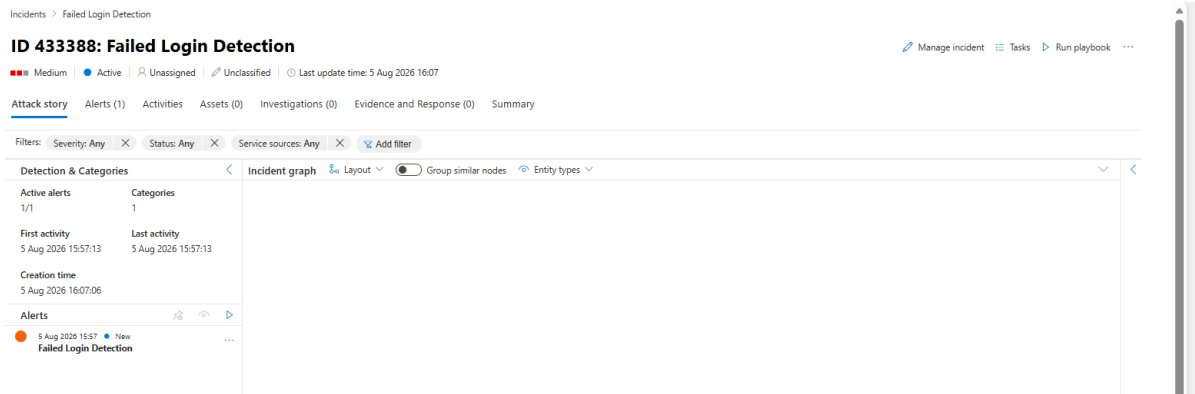
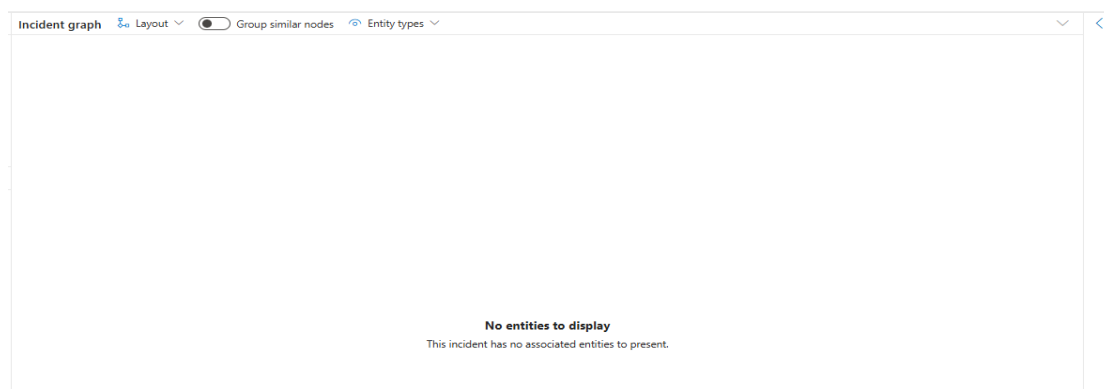


Figure 4.34: Detailed View of Generated Security Incident.

Figure 4.35: Investigation Graph Showing Related Entities.



Incidents > Failed Login Detection

ID 433388: Failed Login Detection Manage incident Tasks Run playbook ...

Medium
Active
Unassigned
Unclassified
Last update time: 5 Aug 2026 16:07

[Attack story](#)
[Alerts \(1\)](#)
[Activities \(2\)](#)
[Assets \(0\)](#)
[Investigations \(0\)](#)
[Evidence and Response \(0\)](#)
[Summary](#)

[Refresh](#)
[Export](#)
[Add comment](#)
2 items
6 Months

[Customize columns](#)

Filter set: [Origin: This incident](#) [Add filter](#) [Reset all](#)

☐ Activity started	☐ Title	☐ Type	☐ Category	☐ Target	☐ Performed by	☐ Trigger	☐ Triggering alert	☐ Activity status	☐ Policy status
☐ 5 Aug 2026 17:10	This is only for testing purposes.	Comment	Comment				Manual		
☐ 5 Aug 2026 16:07	Alert 'Failed Login Detection' was autom...	Alert link	Correlations		Microsoft Defender XDR	Automated		Completed	

Figure 4.36: Analyst Notes and Incident Documentation.

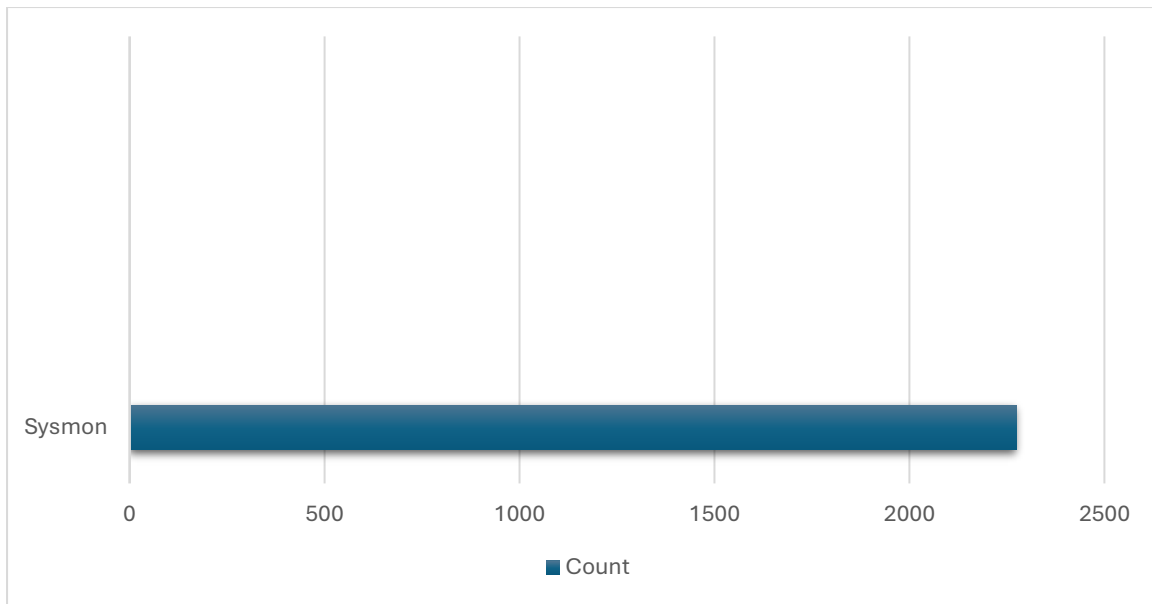
The incident details showed the affected endpoint, detection rule name, timestamp, and severity. This helped validate that the alert was linked to the monitored endpoint and could be reviewed using a structured investigation process.

4.11 Results Analysis.

The implementation results show that the proposed endpoint telemetry-based monitoring system successfully achieved its intended purpose. Endpoint activity was generated on the Windows virtual machine, captured by Sysmon and Windows Event Logs, ingested into Log Analytics Workspace, queried using KQL, and analyzed through Microsoft Sentinel.

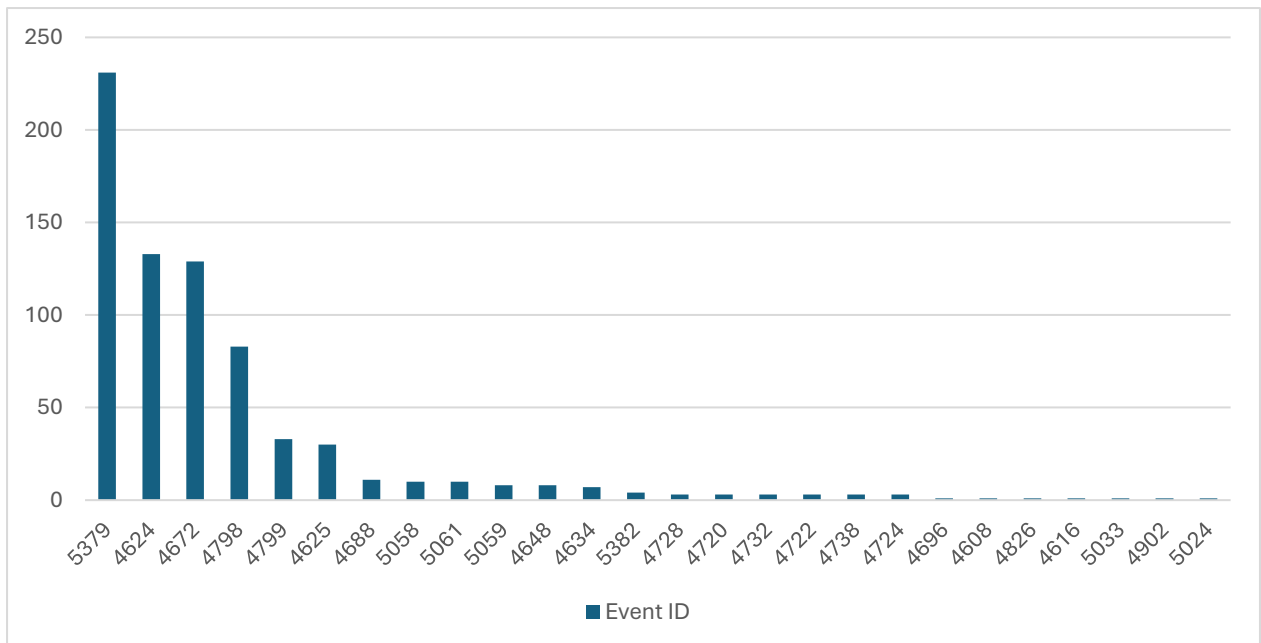
The results confirm that endpoint telemetry can provide valuable visibility into system activity. Queries successfully retrieved security-relevant logs, analytics rules converted selected telemetry into alerts, and incidents provided a structured format for investigation. This demonstrates that endpoint telemetry can support real-time threat detection and incident analysis in an enterprise-like environment.

The analysis also showed that detection effectiveness depends on proper event collection, suitable query logic, and rule tuning. Broad detection rules may generate noise, while narrow rules may miss relevant activity. Therefore, careful tuning is required to improve accuracy and reduce false positives.



Graph 4.1: Total Sysmon Events Collected During Lab Execution.

Graph 4.1 illustrates the total number of Sysmon events collected during the lab execution period. The results indicate that a significant volume of endpoint telemetry was generated and successfully ingested into the monitoring platform, providing sufficient data for threat detection and analysis.



Graph 4.2: Security Events by Event ID.

Graph 4.2 shows the frequency of Windows Security Event IDs collected during the implementation. Event IDs related to authentication, process creation, account management, and privilege usage were observed. The presence of these events confirms that the configured monitoring environment successfully generated security-relevant telemetry suitable for threat detection and incident analysis.

Table 4.7 presents the distribution of incidents generated during the implementation phase based on severity levels configured within Microsoft Sentinel.

Severity	Number of Incidents
Informational	0
Low	5
Medium	7
High	3

Table 4.7: Incidents Count by Severity.

The incident distribution indicates that the implemented analytics rules successfully classified detected activities according to their security relevance. Medium-severity incidents represented the majority of generated findings, while high-severity incidents were associated with activities considered to have greater potential security impact, such as unauthorized account creation. This demonstrates the ability of Microsoft Sentinel analytics rules to prioritize events and support efficient analyst investigation.

4.12 Objective Achievement Analysis.

The implementation was evaluated against the project objectives defined earlier in the dissertation. The results show that the main objectives were successfully achieved. The endpoint environment was configured, telemetry was generated, logs were collected centrally, queries were executed, analytics rules were created, alerts were generated, and incidents were reviewed.

This confirms that the proposed framework is functionally capable of supporting endpoint telemetry-based threat detection and incident analysis in a controlled enterprise-like environment.

Project Objective	Evidence Collected	Status
Configure endpoint lab environment	Azure VM screenshot	Achieved
Generate endpoint telemetry	Sysmon Event Viewer logs	Achieved
Ingest logs into Azure	Log Analytics query output	Achieved
Analyze telemetry using KQL	KQL screenshots	Achieved
Configure analytics rules	Sentinel analytics rule screenshots	Achieved
Generate alerts	Alert screenshot	Achieved
Investigate incidents	Incident details and investigation graph	Achieved

Table 4.8: Objective Achievement Matrix.

4.13 Key Findings.

The following key findings were observed from the implementation:

1. Endpoint telemetry provides detailed visibility into system-level activities such as process execution, authentication events, and system changes.
2. Sysmon improves endpoint monitoring by generating detailed event logs that are useful for security investigation.
3. Centralized log collection through Azure Monitor Agent and Log Analytics Workspace makes endpoint telemetry easier to query and analyze.
4. KQL enables flexible and targeted analysis of collected logs.
5. Microsoft Sentinel supports conversion of detection logic into alerts and incidents.
6. Incident investigation becomes more structured when related alerts, entities, and evidence are available in a centralized SIEM platform.
7. Detection rules require careful tuning to balance visibility and noise reduction.

The findings obtained from the implementation demonstrate that endpoint telemetry provides valuable visibility into system activities that may indicate potential security threats. The integration of Sysmon, Azure Monitor Agent, Log Analytics Workspace, and Microsoft Sentinel enabled effective collection, analysis, and investigation of security events. The results confirm that telemetry-driven monitoring can improve threat detection capabilities and support structured incident response workflows within enterprise environments.

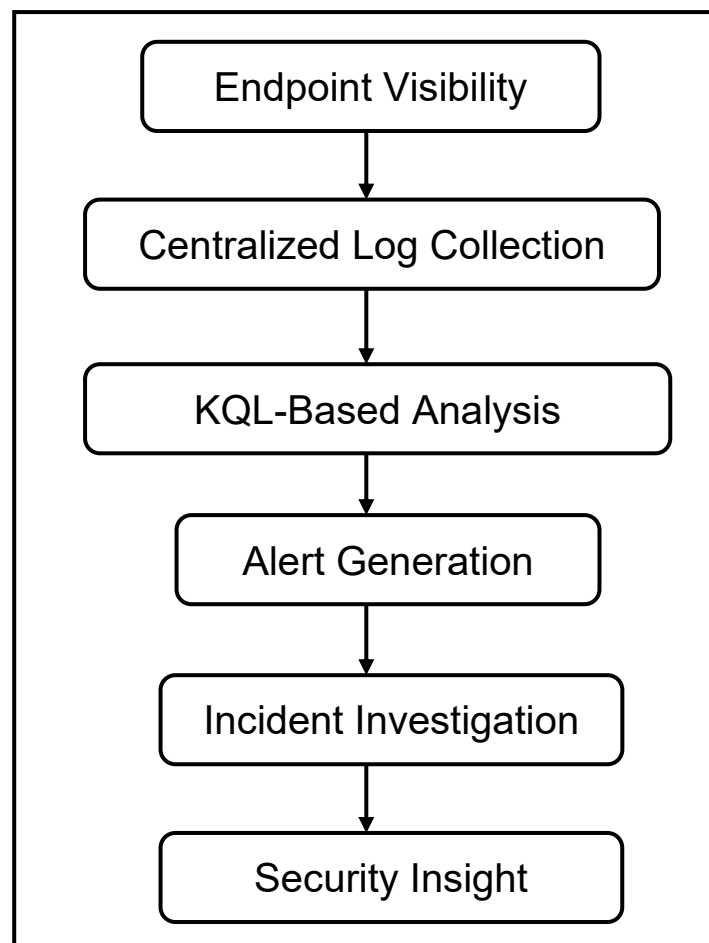


Figure 4.37: Key Finding Model.

4.14 Limitations Observed.

Although the implementation successfully demonstrated endpoint telemetry-based detection, certain limitations were observed. The lab environment had limited event volume compared to a real enterprise environment. The detection logic focused on selected use cases rather than full-scale attack coverage. Some fields depended on logging configuration and were not always available in every query result.

Another limitation was that the system relied mainly on rule-based detection. Rule-based detection is useful for known patterns but requires continuous tuning to reduce false positives and improve accuracy. The implementation also did not include automated response playbooks, which can be considered as future enhancement.

Limitation	Possible Improvement
Limited lab-generated telemetry	Test with larger datasets
Selected detection scenarios only	Add more MITRE ATT&CK mapped detections
Rule tuning required	Improve detection logic based on observed results
Limited automation	Add Sentinel playbooks
Basic visualization	Add dashboards and workbooks
No production data used	Maintain ethical scope but use simulated datasets

Table 4.9: Limitations and Possible Improvements.

4.15 Chapter Summary.

This chapter presented the practical implementation and results analysis of the real-time threat detection and incident analysis system using endpoint telemetry. The implementation included Azure lab setup, Sysmon deployment, telemetry collection, Azure Monitor Agent configuration, Log Analytics validation, Microsoft Sentinel integration, KQL query execution, analytics rule creation, alert generation, and incident investigation.

The results demonstrated that endpoint telemetry can be successfully collected, analyzed, and converted into actionable security alerts and incidents. The implementation also showed that centralized monitoring and SIEM-based investigation can improve visibility and support structured incident analysis in an enterprise-like environment.

Chapter 5

Conclusion and Future Scope

5.1 Introduction.

This chapter presents the conclusion of the dissertation and summarizes the work carried out throughout the project. The dissertation focused on designing and implementing a real-time threat detection and incident analysis framework using endpoint telemetry in an enterprise-like environment. The project combined endpoint monitoring, centralized log collection, SIEM-based detection, and incident investigation into a controlled lab implementation.

The previous chapters discussed the background, literature review, system design, implementation, and results analysis. This chapter highlights the major outcomes, objective achievement, limitations, future enhancements, and conclusion of the work.

5.2 Summary of Work Completed.

The dissertation work began with the identification of cybersecurity challenges in enterprise environments, particularly the difficulty of detecting suspicious activities from large volumes of endpoint logs. The literature review examined endpoint telemetry, SIEM systems, Sysmon, threat detection techniques, Microsoft Sentinel, and incident analysis practices.

Based on the reviewed concepts, a system design was prepared using a Windows endpoint, Sysmon, Azure Monitor Agent, Log Analytics Workspace, and Microsoft Sentinel. The design explained how endpoint activities would be collected, forwarded, analyzed, and converted into alerts and incidents.

The implementation phase successfully demonstrated the practical workflow. Sysmon generated endpoint telemetry, Azure Monitor Agent forwarded selected logs, Log Analytics Workspace stored the data, and Microsoft Sentinel was used for query-based detection, alert generation, and incident investigation.

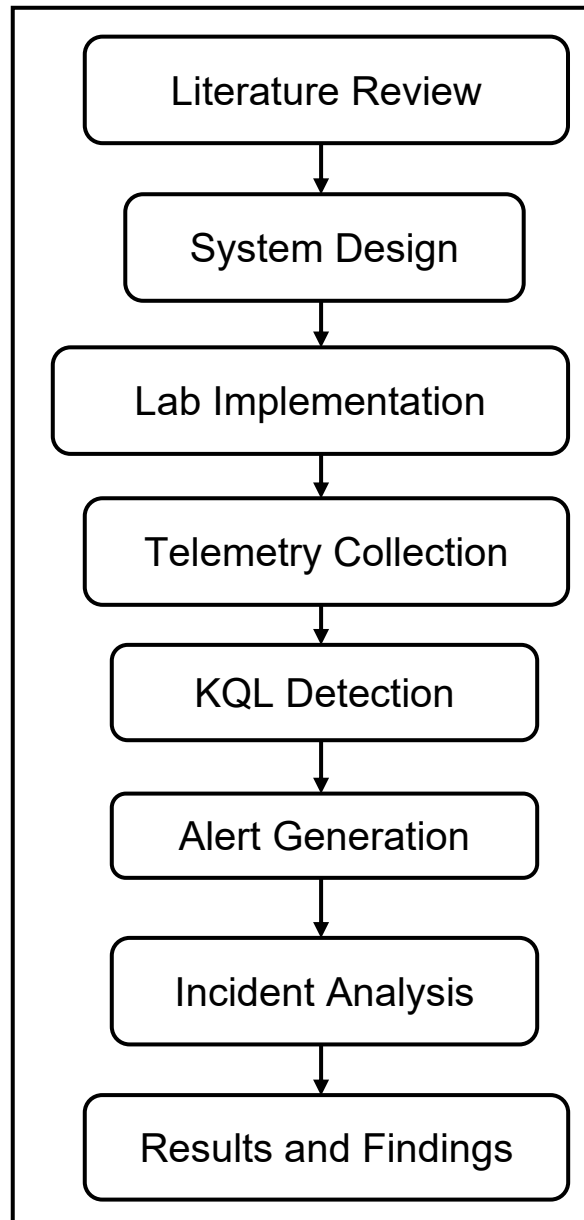


Figure 5.1: Overall Work Completed.

5.3 Achievement of Research Objectives.

The project objectives were successfully achieved through the implementation and validation of the proposed system. The endpoint lab environment was configured, telemetry was generated, logs were collected centrally, detection queries were executed, analytics rules were configured, and incidents were investigated.

The implementation demonstrated that endpoint telemetry can be used as a valuable data source for identifying suspicious activities. It also showed that SIEM integration improves the ability to analyze and investigate security events from a centralized platform.

Research Objective	Achievement Summary
Study endpoint telemetry for threat detection	Achieved through literature review and implementation
Design a telemetry-based monitoring architecture	Achieved through Chapter 3 system design
Collect endpoint logs in a centralized platform	Achieved using Azure Monitor Agent and Log Analytics Workspace
Detect suspicious activities using KQL	Achieved through query execution and analytics rules
Generate alerts and incidents	Achieved through Microsoft Sentinel
Analyze incidents using endpoint evidence	Achieved through Sentinel incident investigation

Table 5.1: Research Objective Achievement Summary.

5.4 Major Findings.

The dissertation produced several important findings. Endpoint telemetry was found to be highly useful for understanding system-level activities that may indicate suspicious behavior. Sysmon provided detailed event visibility that can support detection of process execution, network communication, file activity, and registry changes.

Centralized log collection improved the ability to analyze endpoint activity from a single platform. KQL queries allowed the collected logs to be filtered and investigated efficiently. Microsoft Sentinel provided a structured way to convert detection logic into alerts and incidents, supporting SOC-style investigation.

The results also showed that detection effectiveness depends on the quality of telemetry, configuration of log collection, and tuning of analytics rules. A well-designed detection rule can reduce manual effort, while poorly tuned rules may create unnecessary noise.

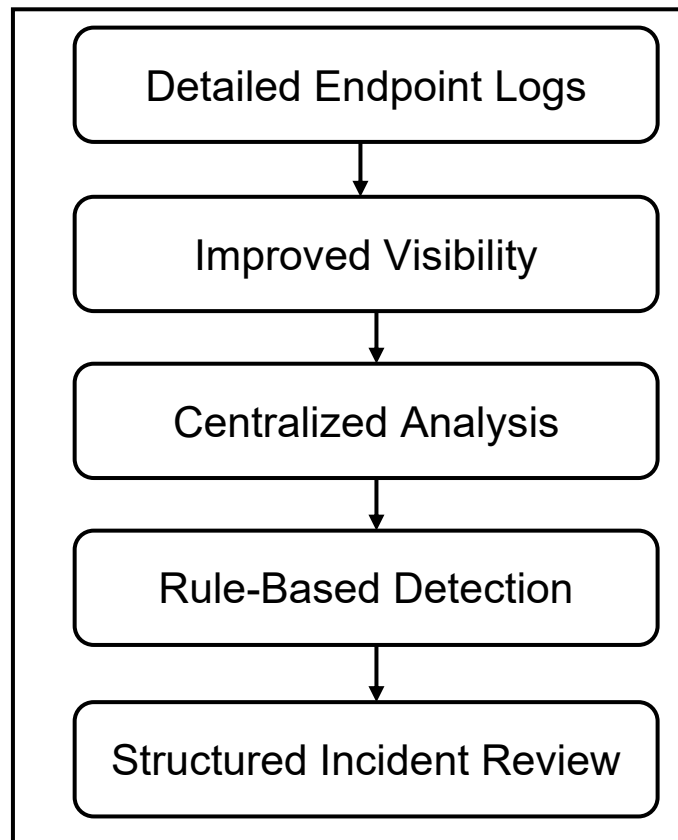


Figure 5.2: Major Findings Summary.

5.5 Limitations of the Work.

The project was implemented in a controlled lab environment and therefore did not include the same scale, complexity, and event volume that would be present in a real enterprise network. The detection scenarios were limited to selected endpoint activities such as PowerShell execution, failed login attempts, process creation, and Sysmon event monitoring.

Another limitation was that the implementation mainly used rule-based detection. While rule-based detection is useful for known suspicious patterns, it requires continuous tuning and may not detect every unknown or advanced attack technique. The project also did not implement automated response actions such as playbooks or endpoint isolation.

Despite these limitations, the project successfully achieved its main objective of demonstrating how endpoint telemetry can support real-time threat detection and incident analysis in a controlled enterprise-like environment.

Limitation	Explanation
Lab-based environment	Results may differ in a large enterprise environment
Limited event volume	Only lab-generated activities were analyzed
Selected detection use cases	Full attack coverage was outside project scope
Manual investigation	Automated response was not implemented
Rule-based detection	Requires tuning to reduce false positives

Table 5.2: Limitations of the Dissertation Work.

5.6 Future Scope.

The current implementation can be extended in several ways to make it more advanced and enterprise ready. Future work may include mapping detection rules to the MITRE ATT&CK framework, adding more detection use cases, creating Microsoft Sentinel workbooks, implementing automated response playbooks, and integrating threat intelligence feeds.

The system can also be improved by testing with larger datasets and simulating more diverse attack scenarios in a safe lab environment. Additional telemetry sources such as firewall logs, identity logs, cloud activity logs, and endpoint protection alerts can also be integrated to improve correlation and detection coverage [13] [17].

Another important future enhancement is automation. Microsoft Sentinel playbooks can be used to automate response actions such as notifying analysts, creating tickets, or triggering containment actions. This would make the system closer to a real SOC environment [10] [13].

5.6.1 Academic Contribution of the Study.

This dissertation contributes to the understanding of endpoint telemetry-based security monitoring by demonstrating a practical implementation of real-time threat detection and incident analysis using Microsoft Azure and Microsoft Sentinel. The study bridges the gap between theoretical cybersecurity concepts and their practical application within a controlled enterprise-like environment.

The implementation highlights how endpoint telemetry can be collected, centralized, analyzed, and converted into actionable alerts and incidents. The findings may assist students, researchers, and security practitioners in understanding the role of telemetry-driven monitoring in modern Security Operations Centres. The study also provides a reproducible implementation approach that can serve as a foundation for future research in SIEM-based threat monitoring and endpoint security analytics.

Future Enhancement	Expected Benefit
MITRE ATT&CK mapping	Improves structured threat understanding
More detection use cases	Increases detection coverage
Sentinel workbooks	Provides better visualization
Automation playbooks	Improves response efficiency
Threat intelligence integration	Enriches alerts with external context
Larger simulation dataset	Improves validation quality
Multi-source log integration	Supports better correlation

Table 5.3: Future Enhancements.

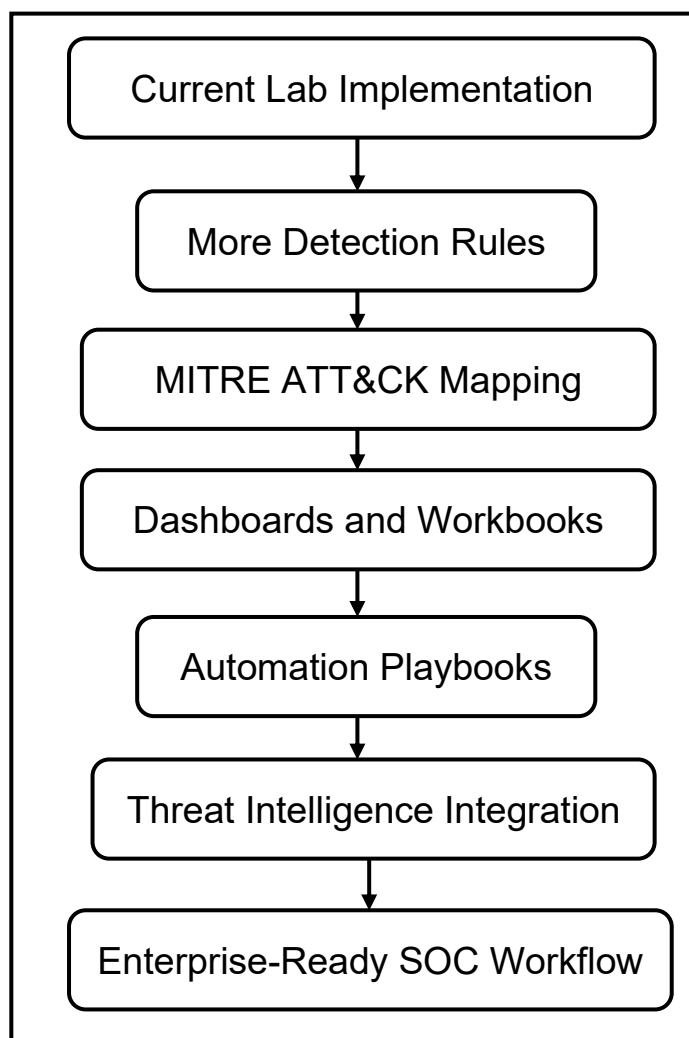


Figure 5.3: Future Scope Roadmap.

This study contributes to cybersecurity research by demonstrating a practical implementation of telemetry-driven security monitoring using cloud-native SIEM technologies. The proposed framework integrates endpoint telemetry collection, centralized log management, rule-based threat detection, alert generation, and incident investigation within a single implementation environment. The project provides a reproducible approach for understanding modern security operations workflows using Microsoft Sentinel.

5.7 Final Conclusion.

This dissertation successfully designed and implemented a real-time threat detection and incident analysis framework using endpoint telemetry. The project demonstrated how endpoint activities can be captured, collected, analyzed, and converted into security alerts and incidents using a SIEM-based workflow.

The implementation showed that Sysmon and Windows Event Logs can provide valuable endpoint telemetry, while Azure Monitor Agent and Log Analytics Workspace can support centralized log collection and querying. Microsoft Sentinel helped convert detection logic into alerts and incidents, enabling structured investigation and analysis.

The project achieved its objective of demonstrating a practical telemetry-driven security monitoring workflow in a controlled enterprise-like lab environment. The findings confirm that endpoint telemetry plays an important role in improving visibility, supporting threat detection, and assisting incident analysis in modern cybersecurity operations.

Overall, the dissertation provides a practical and academically relevant approach to understanding how endpoint telemetry and SIEM technologies can be combined to improve real-time threat detection and incident analysis.

Appendix:

Appendix A: Kusto Query Language (KQL) Queries Used in the Project.

The following Kusto Query Language (KQL) queries were used during the implementation phase to validate telemetry collection, monitor endpoint activities, and investigate security events within Microsoft Sentinel and Log Analytics Workspace.

A.1 Basic Event Validation Query.

```
Event  
| take 20
```

Purpose:

Used to verify that event data was successfully ingested into Log Analytics Workspace and was available for querying.

A.2 Sysmon Event Collection Query.

```
Event  
| where Source contains "Sysmon"  
| project TimeGenerated, Computer, Source, EventID,  
RenderedDescription  
| take 20
```

Purpose:

Used to validate that Sysmon telemetry was being collected and forwarded correctly to the monitoring platform.

A.3 PowerShell Activity Detection Query.

```
Event  
| where RenderedDescription contains "powershell"  
| project TimeGenerated, Computer, EventID, RenderedDescription
```

Purpose:

Used to identify PowerShell execution activity that may indicate administrative actions or potential malicious script execution.

A.4 Failed Authentication Detection Query.

```
SecurityEvent  
| where EventID == 4625  
| project TimeGenerated, Computer, Account, Activity
```

Purpose:

Used to detect failed login attempts and identify possible brute force or credential-guessing activities.

A.5 Process Creation Monitoring Query.

```
SecurityEvent  
| where EventID == 4688  
| project TimeGenerated, Computer, Account, Process,  
CommandLine
```

Purpose:

Used to monitor newly created processes and review command-line activities on the endpoint.

Appendix B: Microsoft Sentinel Analytics Rules Implemented.

The following analytics rules were configured within Microsoft Sentinel to automate detection and alert generation.

Analytics Rule	Detection Purpose	Severity
PowerShell Activity Detection	Monitor PowerShell execution	Medium
Failed Login Detection	Detect authentication failures	Medium
Local User Creation Detection	Detect creation of local accounts	High

These rules converted telemetry data into actionable alerts and security incidents.

Appendix C: Important Security Event Identifiers.

The project monitored selected Sysmon and Windows Security Event IDs relevant to endpoint security monitoring.

Event ID	Source	Description
1	Sysmon	Process Creation
3	Sysmon	Network Connection
11	Sysmon	File Creation
13	Sysmon	Registry Value Modification
22	Sysmon	DNS Query
4625	Windows Security Log	Failed Login Attempt
4688	Windows Security Log	Process Creation

These events were used as primary telemetry sources for threat detection and incident investigation.

Appendix D: Azure Resources Configured During the Implementation.

The project utilized the following Azure services and resources.

Azure Resource	Purpose
Resource Group	Organization of project resources
Windows Virtual Machine	Endpoint telemetry generation
Azure Monitor Agent (AMA)	Telemetry collection and forwarding
Data Collection Rule (DCR)	Definition of collected log sources
Log Analytics Workspace	Centralized log storage and analysis
Microsoft Sentinel	Security Information and Event Management (SIEM) platform

These resources formed the foundation of telemetry collection and security monitoring architecture.

Appendix E: List of Abbreviations.

Abbreviation	Full Form
SIEM	Security Information and Event Management
SOAR	Security Orchestration, Automation and Response
SOC	Security Operations Center
EDR	Endpoint Detection and Response
KQL	Kusto Query Language
AMA	Azure Monitor Agent
DCR	Data Collection Rule
VM	Virtual Machine
DNS	Domain Name System
MITRE	MITRE ATT&CK Framework

ATT&CK	Adversarial Tactics, Techniques and Common Knowledge
Azure	Microsoft Cloud Computing Platform
Sysmon	System Monitor
LAW	Log Analytics Workspace

Appendix F: Project Outcomes Summary.

The project successfully demonstrated the implementation of a telemetry-driven security monitoring framework using Microsoft Azure and Microsoft Sentinel. The key outcomes achieved include:

- Deployment of a Windows-based endpoint monitoring environment.
- Installation and configuration of Sysmon for enhanced telemetry collection.
- Integration of Azure Monitor Agent with Data Collection Rules.
- Centralized log ingestion into Log Analytics Workspace.
- Development and execution of KQL-based detection queries.
- Creation of Microsoft Sentinel analytics rules.
- Generation of security alerts and incidents from endpoint telemetry.
- Investigation of incidents using Microsoft Sentinel investigation tools.
- Mapping of selected detections to the MITRE ATT&CK framework.

The implementation validated that endpoint telemetry provides valuable visibility for threat detection and incident analysis within enterprise environments.

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